

Beyond the Training Distribution: A Survey of Reasoning Generalization in Large Language Models

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Abstract

Large language models (LLMs) have achieved strong performance on mathematical, logical, and scientific reasoning tasks. However, this progress is commonly measured on benchmarks that remain close to the distributions encountered during training. Small changes in notation or problem structure, as well as deeper reasoning or new domains, can lead to substantial performance degradation, raising a central question: *when and how can LLMs generalize their reasoning beyond their training distributions?* This paper presents a comprehensive survey of reasoning generalization in LLMs. We organize existing work into four complementary pillars: (1) training methods that encourage transferable reasoning, (2) inference strategies that adapt computation at test time, (3) architectural designs that support reusable operations, and (4) analyses that characterize when and why generalization succeeds or fails. Building on this taxonomy, we compare the assumptions, evaluation settings, and generalization claims of representative approaches. We further summarize the main empirical findings and theoretical results, and discuss the evaluation practices and open challenges that shape this emerging research direction.

1 Introduction

Recent advances in prompting, post-training, and test-time computation have improved the reasoning capabilities of large language models (LLMs). Models can now solve difficult mathematical problems, construct multi-step proofs, and use external tools. These advances raise a central question: when and how can LLMs generalize their reasoning beyond their training distributions?

Strong benchmark performance does not resolve this question. Accuracy can vary across equivalent numerical versions of a problem and decline when irrelevant clauses are added (Mirzadeh et al., 2025). Models also struggle with deeper or wider proof

trees (Opedal et al., 2025) and unfamiliar combinations of scientific domains (Zhiren et al., 2026). Similar failures occur under symbol remapping, longer inputs, new task compositions, and changed tool interfaces. These findings suggest that success on held-out examples alone does not establish that a model has learned a reusable reasoning procedure.

Assessing reasoning generalization requires a clear account of what is “unseen.” A task absent from supervised fine-tuning may still resemble pre-training data, while extra samples or tool calls may extend system performance without improving the base model. Studies also differ in scope, ranging from evaluations of frontier LLMs to experiments on small transformers and formal results under restricted assumptions. Comparisons therefore need to state what the model has encountered, what changes at test time, and which components remain fixed.

Efforts to improve reasoning generalization mainly span training, inference, and architecture. Training methods seek transferable procedures through data, objectives, and curricula. Inference methods use structured reasoning, additional computation, verification, and external memory to apply a fixed model to unfamiliar tasks. Architectural studies investigate recurrent depth, information routing, and recurrent memory. Analytical studies examine the resulting behavior, internal mechanisms, and theoretical limits. Existing surveys cover broad NLP generalization (Hupkes et al., 2023), out-of-distribution learning (Liu et al., 2021), inductive reasoning (Chen et al., 2025b), or generalization in language-model agents (Zhang et al., 2025b). A synthesis focused on how reasoning procedures transfer across explicit distribution shifts is still needed.

This survey brings these research directions together. As shown in Figure 1, we organize the literature into four pillars: *Training for Generalization* (§3), *Inference for Generalization* (§4), *Ar-*

chitecture for Generalization (§5), and Analysis of Generalization (§6). Together, these pillars organize methods that support reasoning generalization and evidence about the conditions under which it succeeds or fails.

Our key contributions are threefold: (1) **Systematic Taxonomy**. We present a unified taxonomy that organizes studies of reasoning generalization into four pillars (Figure 1), clarifying their scope and objectives. (2) **In-depth Discussion**. Building on this taxonomy, we analyze representative works, comparing training strategies, inference procedures, architectural designs, and behavioral, mechanistic, and theoretical evidence to examine when and how reasoning generalizes. (3) **Challenges and Future Directions**. We identify key challenges in transfer, length and depth extrapolation, and interpretability, and outline promising research directions in transferable training, stable recurrent computation, and verification (§7).

2 Preliminaries

Definitions. Let x denote a reasoning problem, y its answer, and a the inference procedure applied by a model with parameters θ . The resulting system is written as $f_{\theta,a}(x)$, with expected risk $R_P(\theta, a) = \mathbb{E}_{(x,y) \sim P}[\ell(f_{\theta,a}(x), y)]$ on a distribution P . Evaluation is in-distribution (ID) when training and test examples follow the same relevant data-generating process, and out-of-distribution (OOD) when a specified property of that process changes (Liu et al., 2021). When the two losses are comparable, the transfer gap is $R_{\text{OOD}}(\theta, a) - R_{\text{ID}}(\theta, a)$. We use *reasoning generalization* to describe the ability to preserve a reasoning procedure under such a change. This includes compositional generalization to new combinations of familiar primitives (Lake and Baroni, 2018) and length generalization to longer or deeper instances (Anil et al., 2022).

Overview. Following Figure 1, *Training for Generalization* (§3) reviews methods that update model parameters, *Inference for Generalization* (§4) covers methods that operate on a fixed model, and *Architecture for Generalization* (§5) studies designs that support reusable computation. Beyond these methods, *Analysis of Generalization* (§6) examines what transfers and why. Methods spanning several pillars are discussed under their primary source of generalization, and §7 outlines open challenges and future directions.

3 Training for Generalization

Training-based methods update model parameters to improve reasoning generalization under shifts in surface form, composition, length, difficulty, domain, or language. The central challenge is that training can reinforce any pattern that fits the training data, including shortcuts that break under shift (Dziri et al., 2023; Mirzadeh et al., 2025). We therefore organize these methods by training stage and, within each stage, by the component they modify (datasets and protocols are in Appendix C).

3.1 Pre-Training

At the earliest stage, pre-training provides the model with its initial knowledge and reasoning primitives (i.e., basic operations). By the component they modify, methods fall into data curation and optimizer design. Data curation controls what the corpus contains, covering the primitives that later compositions require (Zhang et al., 2025a) and raising the ratio of inferred to stated facts so that the model acquires rules rather than memorizing (Wang et al., 2024a; Abramov et al., 2025). Optimizer design instead changes how the model fits the same corpus, either by aligning gradients across data sources (Chen et al., 2026a) or by using small initialization and strong weight decay to favor primitives over memorization (Zhang et al., 2024d, 2025d), although generalizing solutions reached by training far beyond overfitting can transfer poorly to new knowledge (He et al., 2026).

3.2 Mid-Training

Subsequently, mid-training inserts a stage between pre-training and task-specific adaptation, which narrows the gap between the pre-training and post-training distributions and supplies skills required by the target tasks; without it, a model lacking such a skill tends to memorize the tasks during post-training rather than acquire it (Zhang et al., 2025a; Jin et al., 2025). By the objective of this stage, we distinguish two common approaches: (1) **Continued Pre-Training** methods, which apply next-token prediction to raw text from a domain under-represented in pre-training, directly narrowing this gap (Shao et al., 2024; Fesser et al., 2026). (2) **Intermediate Fine-Tuning** methods, which train on solved problems that require no domain knowledge, such as program-generated deduction or symbolic problems, so that the acquired reasoning transfers to other domains and to realistic prob-

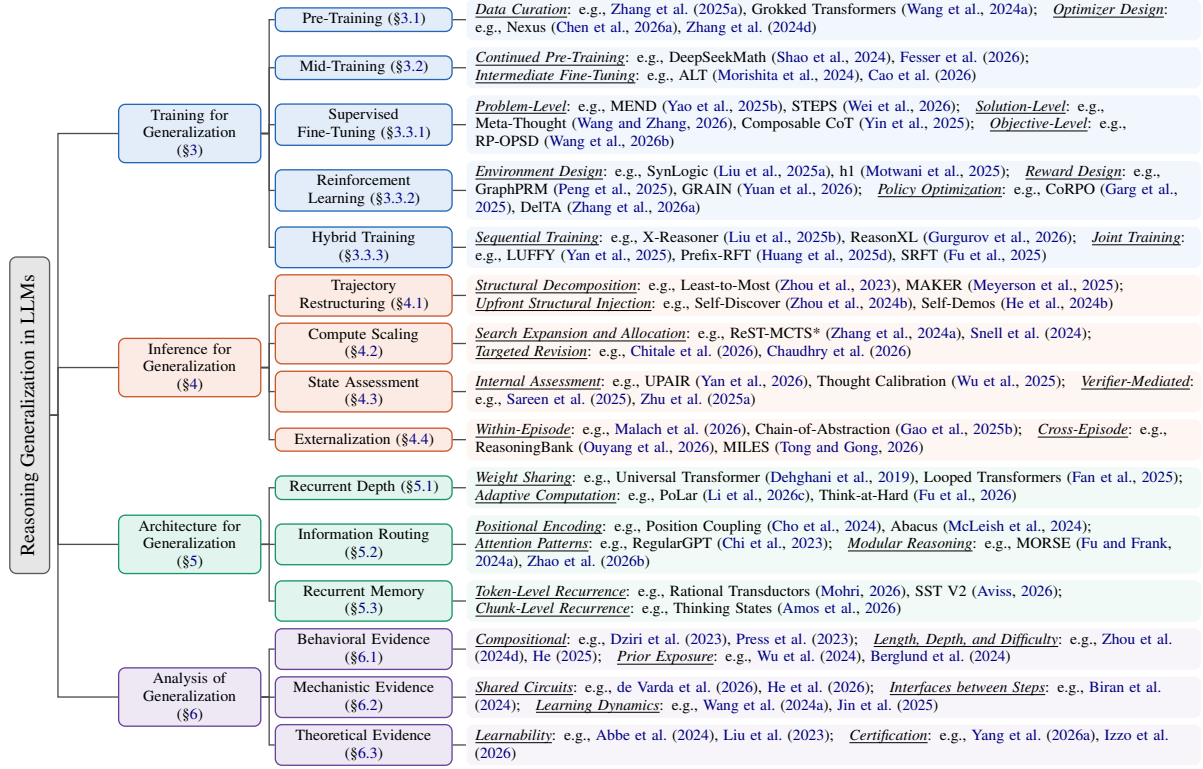


Figure 1: Taxonomy of reasoning generalization in LLMs.

lems in natural language (Morishita et al., 2024; Cao et al., 2026). Beyond such direct transfer, this stage also prepares later training, after which the model generalizes better to OOD tasks and composes separately learned skills into novel combinations (Shrestha et al., 2025; RRV et al., 2026; Cheng et al., 2025a).

3.3 Post-Training

Post-training fine-tunes the model on reasoning tasks and is where the ID/OOD gap most easily opens: a model accurate on training tasks can still fail on rephrased, longer, or out-of-domain problems (Mirzadeh et al., 2025; Opedal et al., 2025; Zhiren et al., 2026). We cover Supervised Fine-Tuning (SFT), Reinforcement Learning (RL), and Hybrid Training.

3.3.1 Supervised Fine-Tuning

SFT trains the model to imitate demonstrations, i.e., step-by-step solutions to training problems, and thus tends to memorize their spurious surface correlations, becoming brittle under shift (Chu et al., 2025; Peng et al., 2026). Accordingly, SFT methods intervene at the level of the problems, the solutions, or the objective.

Problem-Level Methods in SFT. To begin with, this line of work modifies the training problems so that the shift anticipated at test time is already represented in training, leaving no single surface pattern for the model to memorize. (i) **Problem Rewriting** presents each problem in variants that differ only in solution-irrelevant details, teaching the model that the answer stays fixed under surface shift (Yao et al., 2025b; Gui and Ji, 2025). (ii) **Problem Generation** creates problems for regions the original data does not cover, such as rare skill combinations or slightly harder instances, which strengthens compositional and length generalization (Wei et al., 2026; Lee et al., 2025). (iii) **Difficulty Curriculum** orders or selects training problems by difficulty, generalizing to harder or OOD problems (Liu et al., 2026b,d); its benefit nevertheless depends on the shift, as difficulty ordering shows no robust advantage over random sampling for compositional generalization (Mordig et al., 2026).

Solution-Level Methods in SFT. In contrast, these methods rewrite the solutions that the model imitates. The central insight is that imitation generalizes only as far as its target does, so only a target exposing a reusable procedure transfers to new problems (Peng et al., 2026; Matsutani et al., 2026). (i) **Abstract Solutions** remove problem-

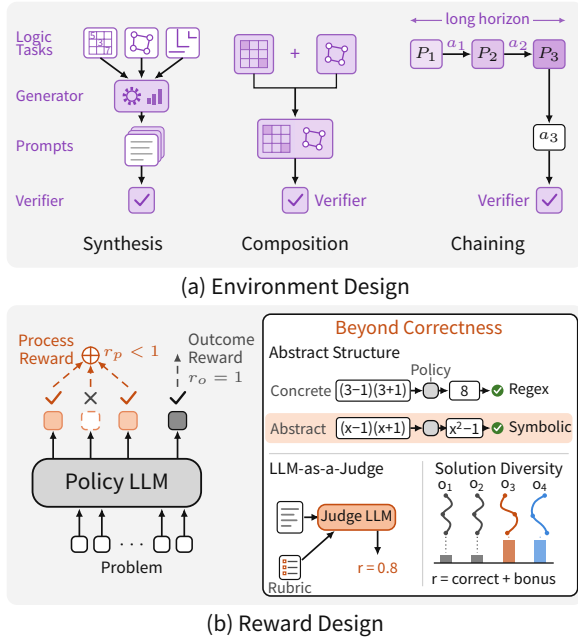


Figure 2: Environment and reward design in RL: (a) building verifiable training problems, and (b) process, outcome, and beyond-correctness rewards.

specific details and teach a general strategy or rule before applying it to a specific problem, addressing surface and domain shift (Wang and Zhang, 2026; Peng et al., 2026). (ii) **Stepwise Solutions** write out every step of a rule allowing it to run for more steps or combine with others and thus to cope with length and compositional shift (Hu et al., 2024; Wang et al., 2025; Hu et al., 2025b; Yin et al., 2025; Hua et al., 2025).

Objective-Level Methods in SFT. Inspired by knowledge distillation (Hinton et al., 2015) and invariance-based regularization from domain generalization (Arjovsky et al., 2019), these methods modify the loss so that what would not hold under shift contributes less to the update. (i) **Teacher-based Objectives** match the output distribution of a teacher on the student’s own solutions instead of imitating a single written solution, which removes the mismatch with the expert solutions seen in training and enables transfer across domains, reasoning horizons, and languages (Li et al., 2026b; Wang et al., 2026b). More selective variants restrict this matching to tokens or spans the student can support, which yields further OOD gains (Jukić and Titov, 2026; Zhao et al., 2026c). (ii) **Regularization-based Objectives** constrain or reweight the imitation loss, keeping only gradient directions shared by logically equivalent problems or reweighting

samples, including wrong-answer ones, and outperform standard SFT on OOD problems (Cheng et al., 2026; Tian et al., 2026). Despite these gains, a teacher given rich privileged information can suppress the student’s uncertainty and degrade accuracy on unseen problems (Kim et al., 2026).

3.3.2 Reinforcement Learning

Unlike SFT, RL replaces reference solutions with a reward, usually whether a checker accepts the final answer, and updates the model toward higher-reward samples (Shao et al., 2024). Without imitation, RL reportedly generalizes to problem variants and domains where SFT degrades (Chu et al., 2025; Huan et al., 2025). Yet RL generalization is bounded: OOD gains seem to appear only when the target task aligns with pre-training (Ni et al., 2025), and accuracy collapses at a few times the training depth (Wang et al., 2026a). We further group RL methods by the component they modify: the environment, the reward, or the policy update (Figures 2 and 3).

Environment Design. First, these methods change the problems and checkers that the model practices on (Figure 2a) and hence control which tasks, difficulties, and reasoning horizons training covers. (i) **Synthesis** employs generators that turn logic tasks into checkable problems of controlled difficulty, with gains that extend to OOD puzzles and other domains (Liu et al., 2025a; Chen et al., 2025a); along this line, even environments built on fictional knowledge or games transfer to real-world problems (Kabra et al., 2026; Feng et al., 2026). (ii) **Composition** merges existing environments into tasks that require several skills at once, generalizing to unseen benchmarks (Xiang et al., 2026). (iii) **Chaining** feeds the answer of each short problem into the next, letting a single verifiable answer span a longer horizon and carry over to longer and harder problems (Motwani et al., 2025).

Reward Design. Second, instead of changing the problems, another family changes what is rewarded (Figure 2b). The central insight is that an outcome-only reward credits any path to a correct answer, including shortcuts that hold only in distribution; thus, this family also rewards properties of the reasoning itself. (i) **Process Rewards** score each intermediate step (r_p), not merely the final answer judged by outcome rewards (r_o), favoring a correct procedure over such shortcuts (Salimi et al., 2026; Peng et al., 2025). (ii) **Rewards beyond**

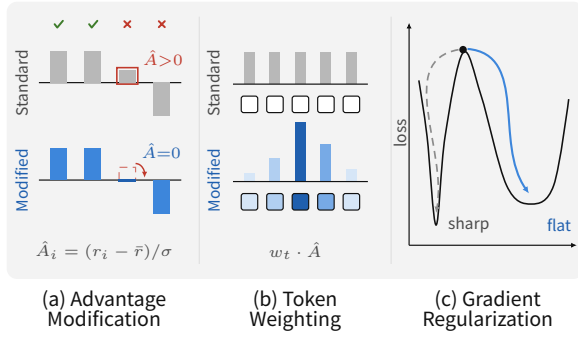


Figure 3: Policy optimization in RL: standard GRPO updates (gray) versus modified ones (blue).

Correctness score properties of a solution other than its answer, in three forms. *Abstract structure* rewards favor solutions that capture the problem symbolically regardless of its names and narrative, narrowing the OOD gap of SFT and increasing robustness to surface perturbations (Yuan et al., 2026; Gao et al., 2025a). An *LLM-as-a-judge* replaces or complements rule-based checkers, extending rewards to answers that rules cannot verify (Ma et al., 2025b; Bhattarai et al., 2026) and raising accuracy outside the training domain (Zha et al., 2025; Ma et al., 2025b; Bhattarai et al., 2026). Lastly, *solution diversity* rewards grant a bonus to solutions that differ from the rest of the group, yielding higher out-of-domain accuracy (Hu et al., 2025c; Zhou et al., 2025; Maniparambil et al., 2026).

Policy Optimization. Finally, rather than changing the problems or the reward, these methods modify the policy update itself, so that it does not reinforce shortcuts that hold only on the training data. Figure 3 illustrates three such modifications to the standard GRPO update (Shao et al., 2024). (i) **Advantage Modification** reshapes the advantage, i.e., how much a rollout outscores its group, ensuring that incorrect rollouts with partial credit never receive a positive update, strengthening OOD transfer (Garg et al., 2025). (ii) **Token Weighting** assigns each token its own credit according to how strongly it signals a correct rather than an incorrect rollout (Zhang et al., 2026a; Xu et al., 2026b), avoiding the OOD degradation of standard updates (Xu et al., 2026b). (iii) **Gradient Regularization** penalizes the gradient norm to steer training toward flatter regions where the reward is more accurate, which curbs reward hacking, i.e., exploiting flaws of the reward (Ackermann et al., 2026).

3.3.3 Hybrid Training

Alternatively, hybrid methods combine demonstrations that introduce procedures the model cannot learn on its own with rewards that recover OOD accuracy reduced during SFT (Jin et al., 2025; Fesser et al., 2026). Depending on how the two signals are scheduled, we distinguish (i) **Sequential Training**, i.e., SFT followed by RL, which transfers reasoning to other domains, modalities, and languages (Liu et al., 2025b; Gurgurov et al., 2026); since excessive SFT leaves RL unable to improve, Rejuvenation (Liu et al., 2026c) partially resets the model toward its base weights before RL, restoring OOD gains. (ii) **Joint Training**, which combines the two signals within one stage, in the sampled solutions (Yan et al., 2025; Huang et al., 2025d) or in the loss (Fu et al., 2025; Lv et al., 2025; Hu et al., 2026; Ma et al., 2025a; Zhu et al., 2026), with several methods reporting higher OOD accuracy than RL alone (Yan et al., 2025; Fu et al., 2025; Lv et al., 2025). This advantage is challenged by Limozin et al. (2026): with corrected baselines, sequential training surpasses these joint methods, including OOD (Appendix C.3, Table 10).

4 Inference for Generalization

Autoregressive generation can compound errors on out-of-distribution (OOD) tasks. Test-time interventions target different parts of the inference process: structuring trajectories (Trajectory Restructuring), exploring alternatives (Compute Scaling), evaluating reasoning reliability (State Assessment), and using external state or tools (Inference-Time Externalization). We group methods by their primary intervention, even when they also serve secondary roles. Their effectiveness remains bounded by model capabilities: inference scaling cannot synthesize algorithmic reasoning entirely absent from training (Tong et al., 2026). Empirical evidence and evaluation caveats are summarized in Table 15, with benchmarks and metrics in Tables 13 and 14.

4.1 Trajectory Restructuring

These methods organize free-form generation without updating model weights. **Structural decomposition** breaks complex reasoning into manageable steps (Zhou et al., 2023; Zhang et al., 2024c), using independently executed micro-agents (Meyerson et al., 2025) or predefined atomic skills (Chen et al., 2024). **Upfront structural injection** constrains generation in advance by restricting valid an-

swers (Ma et al., 2026), constructing task-specific procedures (Zhou et al., 2024b), or synthesizing query-specific demonstrations (He et al., 2024b). The latter addresses out-of-demonstration generalization, which should not be equated with an explicit shift from the pretraining distribution.

4.2 Compute Scaling

When one trajectory is insufficient, extra test-time compute can explore alternatives or revise errors. **Search expansion and allocation** uses process-level signals to guide exploration: ReST-MCTS* (Zhang et al., 2024a) combines a Process Reward Model (PRM) with Monte Carlo Tree Search (MCTS) to score partial solutions, while MCTS can also improve policy and reward models for finer inference guidance (Chan et al., 2025). Adaptive methods allocate compute according to estimated problem difficulty (Snell et al., 2024), although this alone does not establish OOD generalization. **Targeted revision** concentrates computation on faulty reasoning steps rather than regenerating an entire solution (Chitale et al., 2026).

4.3 State Assessment

State assessment estimates the reliability of intermediate reasoning to guide selection or further computation. **Internal assessment** draws on residual-stream trajectories (Damirchi et al., 2026), uncertainty-progress alignment (Yan et al., 2026), or thought calibration for risk-controlled stopping (Wu et al., 2025). Online calibration can further update reliability estimates under distribution shift (Zhou et al., 2026). **Verifier-mediated assessment** uses auxiliary models for candidate selection or adaptation: joint reasoner-verifier training supports inference-time selection in evaluated mathematical and physics tasks (Sareen et al., 2025), while verifier-guided test-time training adapts the generator using high-confidence outputs on unlabeled target data (Moradi et al., 2025). However, PRMs can generalize poorly to harder problems or solutions from different models (Zhu et al., 2025a).

4.4 Inference-Time Externalization

Externalization gives inference access to working state or computation beyond the model’s internal state. **Within-episode externalization** uses interactive memory to support length extrapolation (Malach et al., 2026), separates abstract reasoning from tool-supplied values (Gao et al., 2025b), and supports tool selection or restraint (Alazraki

and Rei, 2025; Qian et al., 2025). These gains do not by themselves establish that the model generalizes without external resources. **Cross-episode externalization** stores strategies distilled from prior successes and failures in persistent memory (Ouyang et al., 2026); modular sub-instructions and learned selection mechanisms support memory retrieval (Tong and Gong, 2026), although cross-domain transfer is not directly established by the main evaluations.

5 Architecture for Generalization

Architectural approaches change the computation available to a model: how it reuses transformations, connects representations, or carries state forward. Such changes provide inductive biases that can support reasoning generalization by allowing a model to reuse learned operations when problems grow longer or combine familiar steps in new ways.

5.1 Recurrent Depth

Recurrent depth repeatedly applies a shared transformation, increasing computation without a separate parameter set at each depth (Geiping et al., 2026; Zeng et al., 2025a; Zhu et al., 2025c). This can support length generalization when longer problems require more iterations of the same operation, as shown on controlled algorithmic tasks (Fan et al., 2025). Designs differ in which computation is reused and how its iteration count is adjusted across inputs or tokens. **Weight Sharing.** Shared recurrent blocks support generalization to deeper compositions of facts (Kohli et al., 2026). On computation graphs, recurrence combined with latent supervision, discretization, and error correction enables generalization to larger instances (Altabaa et al., 2025). In pretrained language models, a shared middle block supports depth extrapolation on controlled rule-following tasks, although zero-shot transfer to verbal formulations remains limited (Shapiro, 2026). Looped language models also improve compositional tool calling involving dependent calls (Popescu et al., 2026). **Adaptive Computation.** PoLar (Li et al., 2026c) learns input-dependent programs that skip or repeat groups of layers in a frozen backbone. Think-at-Hard (Fu et al., 2026) selects tokens for extra latent iterations, and its math-only variant also improves on MMLU-STEM. Extra iterations can still degrade predictions when recurrent states drift, limiting depth extrapolation (Viakhirev et al., 2026).

5.2 Information Routing

Information routing determines how tokens are addressed, which dependencies attention can access, and which components process them (Vaswani et al., 2017; Shazeer et al., 2017). For reasoning generalization, these designs aim to preserve the connections needed by a computation as input length or composition changes. **Positional Encoding.** Position Coupling (Cho et al., 2024) assigns shared position identifiers to digits of equal significance across operands, while Abacus embeddings (McLeish et al., 2024) encode digit positions within each number. Structural symmetry and alternative positional representations also improve arithmetic extrapolation (Sabbaghi et al., 2024; Shen et al., 2023). Multi-level position coupling (Cho et al., 2025) combines multiple position coordinates with task-specific scratchpads to generalize across operand lengths and counts. TAPE (Zhu et al., 2025b) instead updates positional representations across layers using sequence context, improving arithmetic length extrapolation. Much of this arithmetic evidence comes from specialized models. **Attention Patterns.** RegularGPT (Chi et al., 2023) combines sliding-dilated attention with shared layers and length-dependent depth, composing local results over larger spans to recognize regular languages beyond training lengths. Attention Bias Calibration (Duan et al., 2023) converts attention patterns learned on shorter inputs into biases that support arithmetic length extrapolation. **Modular Reasoning.** MORSE (Fu and Frank, 2024a) learns input-dependent masks that route tokens to specialized attention heads, improving generalization across entailment-tree lengths and shapes. In sparse expert language models, more complex skill compositions benefit from more active experts, while excessive activation can hurt simpler compositions (Zhao et al., 2026b).

5.3 Recurrent Memory

Recurrent memory carries an internal state across computation steps, allowing later steps to reuse earlier information (Bulatov et al., 2022). Updates occur across tokens or reasoning chunks, and generalization depends on retaining the needed information as sequences grow. **Token-Level Recurrence.** Rational Transductors (Mohri, 2026) combine weighted-automaton recurrence with transformer attention, providing theoretical and synthetic-task evidence for length generalization. SST V2 (Avis,

2026) blends a carried state into pretrained feed-forward layers and reports gains beyond math fine-tuning, although its aggregate scores count a question as solved if any tested iteration depth succeeds. **Chunk-Level Recurrence.** Thinking States (Amos et al., 2026) compresses intermediate thoughts into a fixed-size state for later chunks, improving length extrapolation on parity and variable-tracking tasks, although the model still accesses preceding context alongside this state.

6 Analysis of Generalization

While the methods in §3–§5 aim to improve performance under distribution shift, analytical studies examine what actually transfers and why. We organize this work by three complementary types of evidence: behavioral evidence measures performance under controlled shifts; mechanistic evidence probes internal computation, mostly in small models; and theoretical evidence characterizes what can be represented, learned, and certified. Table 16 summarizes each line of analysis, with details in Appendix D.

6.1 Behavioral Evidence

Compositional generalization. Accuracy declines as computation graphs grow deeper and wider (Dziri et al., 2023), models can fail to chain two facts they both know (Zhang et al., 2026b), and this gap does not shrink with scale in the GPT-3 family (Press et al., 2023). **Length, depth, and difficulty.** Transformers fail on nesting deeper than seen in training (He, 2025), short-to-long transfer varies across random seeds (Zhou et al., 2024d), and training only on easy or only on hard examples yields inconsistent gains (Kordi et al., 2026). **Prior exposure.** Whether a test is “unseen” depends on the training stage it is compared against: accuracy drops on familiar tasks posed in an unfamiliar number base (Wu et al., 2024), facts learned as “A is B” are not retrieved as “B is A” (Berglund et al., 2024), and RL transfer requires sufficient base-model competence (Lu et al., 2025).

6.2 Mechanistic Evidence

Mechanistic studies reveal how internal computations shape generalization. **Shared circuits.** If two problem variants are solved by the same procedure, their internal computations should overlap. de Varda et al. (2026) show that, in LLM arithmetic, the overlap between circuits for numeric

and verbal formats predicts which formats are hardest, which models transfer best, and which items are solved. However, such a circuit can transfer poorly once it must use newly integrated knowledge (He et al., 2026). **Interfaces between steps.** Multi-step reasoning can also fail when an intermediate result does not reach the next step. Biran et al. (2024) find that LLMs resolve the bridge entity of a two-hop query in early layers but begin the second hop in later layers that may lack the required knowledge; patching a later hidden state into an earlier layer, chosen per query, recovers up to 66% of failed queries. **Learning dynamics.** When several solutions fit the training data, optimization determines which one the model reaches. Grokked transformers generalize OOD on comparison but not on composition (Wang et al., 2024a). In LLM post-training, OOD performance can peak early in SFT and then decline while ID performance still improves (Jin et al., 2025), making the evaluated checkpoint part of the generalization claim.

6.3 Theoretical Evidence

Theoretical work distinguishes three properties that empirical results often conflate: *expressivity*, whether a model class can represent a procedure that generalizes in length or composition; *learnability*, whether finite training selects this procedure over shortcuts that fit the same data; and *certification*, whether a finite test can establish that it extrapolates. Each result holds only under its stated encoding, precision, and computation budget. **Learnability.** Two conditions govern whether training selects the extrapolating procedure. The first is locality: Abbe et al. (2024) show, theoretically under additional assumptions and empirically, that targets with a high globality degree are not efficiently learnable by regular transformers, whereas inductive scratchpads, which condition each step only on the question and the previous state, break this barrier and improve length transfer. The second is inductive bias: Liu et al. (2023) prove that $O(\log T)$ layers suffice to simulate any finite automaton on inputs of length T , yet standard training in their experiments finds shortcuts that are brittle outside the training distribution. **Certification.** Yang et al. (2026a) prove that no computable length generalization bound exists for C-RASP, even with two layers, so no computable test length certifies the class. Positive results hold only for narrower classes: one- and two-layer transformers provably length-generalize when behavior on

longer sequences can be simulated by behavior on shorter training sequences (Izzo et al., 2026).

7 Challenges and Future Directions

Challenges (1) **Limited transfer.** Models that perform well on familiar tasks can fail when known reasoning steps are combined in new ways or applied to different domains (§6.1). Learning procedures that remain effective under these changes is still a major challenge. (2) **Length and depth extrapolation.** Accuracy can decline when problems require more reasoning steps than those seen during training (Wang et al., 2026a). Additional computation at inference does not always resolve this difficulty, as recurrent states may drift (Viakhirev et al., 2026). (3) **Interpretability.** Behavioral scores alone do not show whether a model reuses the same reasoning procedure across tasks. Many mechanistic studies use small models, and their findings may not fully explain transfer in frontier LLMs (§6.2).

Future Directions (1) **Transferable training.** Developing training methods that teach reusable operations is central to improving generalization. Supervision should emphasize the reasoning steps that remain useful when familiar operations are combined in new ways (§3). (2) **Stable recurrent architectures.** Developing recurrent architectures with stable state updates may improve reasoning on longer and deeper problems. These architectures should maintain reliable computation as the number of reasoning steps increases (§5). (3) **Interpretability and verification.** Developing methods to identify and verify reusable computations is important for understanding reasoning generalization. Applying these methods to larger models would help connect mechanistic findings with the failures observed under distribution shift (§6.2).

8 Conclusion

This survey reviews reasoning generalization in LLMs from four perspectives: training, inference, architecture, and analysis. Existing studies show that generalization depends on what models learn, the computation available at inference, and the type of distribution shift. Although current methods achieve promising results in specific settings, none consistently generalizes across different shifts. Future research should develop clearer evaluation protocols and better explain when and why reasoning generalizes.

Limitations

While this survey offers a comprehensive review of existing methods and analyses of reasoning generalization in LLMs, a few limitations remain in literature coverage, comparison across studies, and applicability to larger models. Given the broad scope and rapid growth of this field, our review may have missed some relevant studies. Our focus on explicit reasoning-related distribution shifts limits its coverage of related work and leaves relatively few inference studies. Differences in models, tasks, metrics, and compute restrict the synthesis to qualitative comparisons. Many mechanistic and theoretical findings rely on small models or idealized assumptions, so their relevance to larger models remains uncertain. Finally, our taxonomy assigns each study a primary pillar, although some methods span several.

Ethical considerations

This work surveys publicly available research on reasoning generalization in LLMs and does not involve human participants, collect new personal data, or train or release new models. Reasoning methods may be used for both beneficial and harmful purposes, and improved transfer on selected benchmarks does not establish that a model is safe or reliable in deployment. We therefore encourage assessing generalization alongside bias, privacy, and misuse risks, especially in applications where reasoning errors can harm users.

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1608	<i>Preprint</i> , arXiv:2603.09678.	Julius Steiglechner, Lucas Mahler, and Gabriele	1664
1609	Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarsz,	Lohmann. 2026. LLMs can see the smoke but not the	1665
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1611	Dean. 2017. Outrageously large neural networks:	<i>Preprint</i> , arXiv:2607.12733.	1667
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1614	Shlok Shelat, Jay Raval, Souvik Roy, and Manas Gaur.	verifiable rewards . <i>Preprint</i> , arXiv:2505.24760.	1670
1615	2026. Beyond memorization: Testing LLM reason-	Yiyiyou Sun, Shawn Hu, Georgia Zhou, Ken Zheng, Han-	1671
1616	ing on unseen theory of computation tasks . <i>Preprint</i> ,	naneh Hajishirzi, Nouha Dziri, and Dawn Song. 2025.	1672
1617	arXiv:2601.13392.	OMEGA: Can LLMs reason outside the box in math?	1673
1618	Ruoqi Shen, Sébastien Bubeck, Ronen Eldan, Yin Tat	evaluating exploratory, compositional, and transfor-	1674
1619	Lee, Yuanzhi Li, and Yi Zhang. 2023. Posi-	mative generalization . <i>Preprint</i> , arXiv:2506.18880.	1675
1620	tional description matters for transformers arithmetic .		
1621	<i>Preprint</i> , arXiv:2311.14737.		

1676	Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc V. Le, Ed H. Chi, Denny Zhou, and Jason Wei. 2023. Challenging BIG-Bench tasks and whether chain-of-thought can solve them. In <i>Findings of the Association for Computational Linguistics: ACL 2023</i> .	1730
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1703	Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need . In <i>Advances in Neural Information Processing Systems</i> , volume 30.	1758
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1708	Petar Veličković, Adrià Puigdomènech Badia, David Budden, Razvan Pascanu, Andrea Banino, Misha Dashevskiy, Raia Hadsell, and Charles Blundell. 2022. The CLRS algorithmic reasoning benchmark. In <i>International Conference on Machine Learning (ICML)</i> .	1763
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1714	Ivan Viakhirev, Kirill Borodin, Amirah Almutairi, Serguei Barannikov, Maxim Abramov, and Grach Mkrtchian. 2026. Think shallow, solve deep: Controlling recurrent dynamics for reliable Test-Time depth . <i>Preprint</i> , arXiv:2608.18222.	1769
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1719	Boshi Wang, Xiang Yue, Yu Su, and Huan Sun. 2024a. Grokked transformers are implicit reasoners: A mechanistic journey to the edge of generalization . In <i>Advances in Neural Information Processing Systems 37 (NeurIPS)</i> .	1774
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	Menghua Wu, Cai Zhou, Stephen Bates, and Tommi Jaakkola. 2025. Thought calibration: Efficient and confident test-time scaling . <i>arXiv preprint arXiv:2505.18404</i> .	
	Xinyi Wu, Siyuan Liu, and Ali Jadbabaie. 2026. How data shapes RoPE frequency usage: From positional scale matching to length generalization . <i>Preprint</i> , arXiv:2607.07678.	
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1898	ing Systems , volume 37.	<i>of Machine Learning Research</i> , pages 1049–1071.	1951
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1899	Hugh Zhang, Jeff Da, Dean Lee, Vaughn Robinson,	Ziqi Zhao, Xinyu Ma, Liu Yang, Yujie Feng, Daiting	1953
1900	Catherine Wu, Will Song, Tiffany Zhao, Pranav Raja,	Shi, Jingzhou He, Xin Xin, Zhaochun Ren, and Xiao-	1954
1901	Dylan Slack, Qin Lyu, Sean Hendryx, Russell Ka-	Ming Wu. 2026c. Beyond imitation: Reflective on-	1955
1902	plan, Michele Lunati, and Summer Yue. 2024b. A	policy self-distillation for LLM reasoning . <i>Preprint</i> ,	1956
1903	careful examination of large language model per-	arXiv:2605.28014.	1957
1904	formance on grade school arithmetic . <i>Preprint</i> ,		
1905	arXiv:2405.00332.		
1906	Kaiyi Zhang, Wei Wu, and Yankai Lin. 2026a. DelTA:	Huaixiu Steven Zheng, Xinyun Chen, Minmin Chen,	1958
1907	Discriminative token credit assignment for reinforce-	Azade Nova, Le Hou, Swaroop Mishra, Heng-Tze	1959
1908	ment learning from verifiable rewards . <i>Preprint</i> ,	Cheng, Quoc V. Le, Ed H. Chi, Denny Zhou, and	1960
1909	arXiv:2605.21467.	Hugh Zhang. 2024. Natural plan: Benchmark-	1961
		ing LLMs on natural language planning . <i>Preprint</i> ,	1962
		arXiv:2406.04520.	1963
1910	Minxing Zhang, Yi Yang, Roy Xie, Bhuwan Dhingra,	Gong Zhiren, Tiantong Wu, Jiaming Zhang, Fuyao	1964
1911	Shuyan Zhou, and Jian Pei. 2025b. Generalizability	Zhang, Che Wang, Yurong Hao, Yikun Hou,	1965
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1915	Jin. 2025c. Finite state automata inside transformers	Dimensional scientific knowledge composition .	1969
1916	with Chain-of-Thought: A mechanistic study on state	<i>Preprint</i> , arXiv:2605.14754.	1970
1917	tracking . <i>Preprint</i> , arXiv:2502.20129.		
1918	Yizhou Zhang, Lun Du, Defu Cao, Qiang Fu, and Yan	Cai Zhou, Zekai Wang, Menghua Wu, Qianyu Julie Zhu	1971
1919	Liu. 2024c. An examination on the effectiveness	employment, Flora C. Shi, Chenyu Wang, Ashia	1972
1920	of divide-and-conquer prompting in large language	Wilson, Tommi Jaakkola, and Stephen Bates. 2026.	1973
1921	models . <i>Preprint</i> , arXiv:2402.05359.	Online reasoning calibration: Test-time training en-	1974
		ables generalizable conformal LLM reasoning . <i>arXiv</i>	1975
		<i>preprint</i> arXiv:2604.01170.	1976
1922	Zhongwang Zhang, Pengxiao Lin, Zhiwei Wang, Yaoyu	Denny Zhou, Nathanael Schärli, Le Hou, Jason Wei,	1977
1923	Zhang, and Zhi-Qin John Xu. 2024d. Initializa-	Nathan Scales, Xuezhi Wang, Dale Schuurmans,	1978
1924	tion is critical to whether Transformers fit compos-	Claire Cui, Olivier Bousquet, Quoc Le, and Ed Chi.	1979
1925	ite functions by reasoning or memorizing . <i>Preprint</i> ,	2023. Least-to-Most prompting enables complex rea-	1980
1926	arXiv:2405.05409.	soning in large language models . In <i>International</i>	1981
1927	Zhongwang Zhang, Pengxiao Lin, Zhiwei Wang, Yaoyu	<i>Conference on Learning Representations (ICLR)</i> .	1982
1928	Zhang, and Zhi-Qin John Xu. 2025d. Complex-	Hattie Zhou, Arwen Bradley, Etai Littwin, Noam Razin,	1983
1929	ity control facilitates Reasoning-Based composi-	Omid Saremi, Josh Susskind, Samy Bengio, and Pree-	1984
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1931	arXiv:2501.08537.	formers learn? a study in length generalization . In	1986
1932	Zili Zhang, Yilin Wang, Heng Wang, Herun Wan, and	<i>International Conference on Learning Representa-</i>	1987
1933	Minnan Luo. 2026b. Why knowing both hops is not	<i>tions (ICLR)</i> .	1988
1934	enough: Understanding Two-Hop generalization in	Pei Zhou, Jay Pujara, Xiang Ren, Xinyun Chen, Heng-	1989
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1937	Bohan Jiang, Yancheng Wang, Yingzhen Yang, and	Self-Discover: Large language models self-compose	1992
1938	Huan Liu. 2025. Is Chain-of-Thought reasoning of	reasoning structures . In <i>Advances in Neural Informa-</i>	1993
1939	LLMs a mirage? a data distribution lens . <i>Preprint</i> ,	<i>tion Processing Systems 37 (NeurIPS)</i> , volume 37,	1994
1940	arXiv:2508.01191.	pages 126032–126058.	1995
1941	Guangxiang Zhao et al. 2026a. ScienceArena: Bench-	Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou,	1996
1942	marking LLMs on latest scientific olympiad competi-	Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue	1997
1943	tions . <i>Preprint</i> , arXiv:2608.30517.	Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Gra-	1998
1944	Jinze Zhao, Peihao Wang, Junjie Yang, Ruisi Cai,	ham Neubig. 2024c. WebArena: A realistic web	1999
1945	Gaowen Liu, Jayanth Srinivasa, Ramana Rao Kom-	environment for building autonomous agents . In <i>The</i>	2000
1946	pella, Yingbin Liang, and Zhangyang Wang. 2026b.	<i>Twelfth International Conference on Learning Repre-</i>	2001
		<i>sentations</i> .	2002

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A Literature Collection and Corpus

A.1 Search and Screening

We searched for work on reasoning generalization, OOD reasoning, systematic and compositional generalization, length and depth generalization, reasoning transfer, and test-time generalization. To find papers whose titles did not mention generalization, we also searched abstracts by pairing training-stage terms (pre-training, mid-training, SFT, distillation, RL, RLVR, and self-improvement) with terms such as transfer, unseen composition, symbolic variation, longer input, deeper proof, and OOD evaluation. We ran the queries separately for each submission period so that the much larger 2026 literature would not crowd out relevant work from 2024 and 2025. After deduplication, we screened 6,007 arXiv records and checked related entries in the ACL Anthology and OpenReview. We read every retained abstract and consulted the full paper

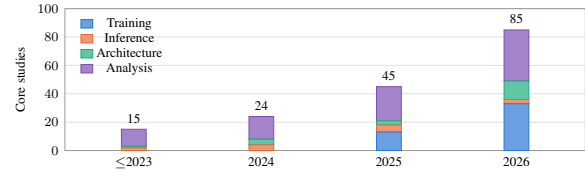


Figure 4: Core studies by primary pillar and year of first arXiv posting ($n = 169$). Bar labels give yearly totals.

whenever the abstract did not state the evaluation split clearly.

We included a core study when it specified the reasoning task, the relevant reference exposure, the test-time shift, and evidence of transfer or failure. Reference exposure may concern model training, in-context demonstrations, or data used to fit an inference component such as a verifier. We excluded work that reported only an average improvement in reasoning scores, broad robustness studies in which reasoning was one of many unrelated capabilities, model-to-model transfer without a shift to unfamiliar problems, and expressivity results without a generalization claim. When a paper used the term “generalization” without stating what changed between reference and target conditions, we examined its experimental setup and data construction before deciding whether to include it. We also discuss contextual methods that explain the design space but do not meet this rule; these are excluded from the core counts.

A.2 Corpus Composition

The corpus contains 169 core studies, each counted once under its primary pillar (Figure 4 and Table 1). Analysis is the largest pillar with 88 studies, followed by training with 46 (38 of them on post-training), architecture with 21, and inference with 14. Most studies (130) first appeared in 2025 or 2026, reflecting the recent growth of the area, although many results remain preprints.

The inference count covers the 13 studies marked † in Table 15 and test-time hinting (Hou et al., 2026), which evaluates transfer to unseen benchmarks. The other 11 entries in the evidence table are contextual methods with indirect or absent shift evidence, including DaC, Self-Discover, compute-optimal scaling, and TTEL; they are excluded from the core counts. Two related scaling studies (Li et al., 2026a; Chaudhry et al., 2026) are counted under training, as their interventions update model parameters. Thus, the number of methods discussed in the inference pillar exceeds

its core-study count without counting a study twice.

Figure 1 shows representative works for each branch, including contextual methods. Table 2 lists additional core studies that the main text does not discuss, under their primary branch. Related surveys and contextual studies are excluded from the corpus counts.

B Synthesis and Evaluation Framework

B.1 Connections Across Pillars

The four pillars describe different parts of one process. Training supplies reasoning primitives and determines which of several fitting procedures the model prefers (§3). Architecture constrains which computations can be represented and repeated efficiently (§5). Inference controls how a learned procedure is elicited, searched, checked, or supplemented with external resources (§4). Analysis tests whether the resulting behavior survives the intended shift and whether the proposed explanation holds (§6).

These parts interact, so improvements in one can remain limited by weaknesses in another. Intermediate reasoning steps help when they expose a reusable computation: in tiny transformers, the gain from traces comes from their content rather than from the extra tokens (Valle and Reid, 2026), and scratchpads improve length transfer when each step depends only on the question and the previous state (Abbe et al., 2024). Recurrent depth helps only while its update remains stable, since recurrent states can drift as the number of iterations grows (Viakhirev et al., 2026). Tools can extend what a system solves without changing what the model has learned. With access to external memory, state-space models extrapolate to inputs far longer than those seen in training (Malach et al., 2026). Without external memory, the same models remain limited by their bounded internal state.

Training and inference also constrain each other. Test-time scaling cannot produce reasoning that is entirely absent from the training distribution (Tong et al., 2026), and a model trained under one decoding procedure may not cooperate with the controller used at deployment. Placing inference-time controllers in the training loop improves transfer to new inference workflows compared with training for a single controller (Choudhury et al., 2026).

Tables 3 and 4 summarize each branch of the training and inference pillars. The training table gives the mechanism through which a branch is

expected to support transfer, the strongest current evidence, and the main risk. The inference table gives the conditions under which a branch helps or fails and the control needed before a gain can be read as transfer.

B.2 Evaluation Framework

Claim boundaries. A generalization result should specify which components are evaluated and held fixed, as gains enabled by prompting, search, verification, or memory need not hold for the model alone. Generation must also be distinguished from selection: greedy accuracy shows what a model produces reliably, large-pass@ k accuracy what it can produce occasionally, and verifier-selected accuracy what a larger system can recover. Gains on the matched target distribution show specialization, whereas gains under held-out shifts provide evidence of reuse. Stating these conditions makes comparisons clearer without treating every gain as evidence of the same capability.

Elements of a claim. Table 5 lists the minimum information needed to place a result within these boundaries. The test shift needs to be constructed so that its effect can be attributed: each OOD condition has a matched ID control, one factor changes at a time where possible, and the suite includes both surface and structural changes (Table 7). Explanatory claims need controls of their own. A proposed mechanism should be tested with counterfactual rules or perturbations across several task families, training seeds, and checkpoints, and a causal intervention should be evaluated on the split that established the behavioral effect, because a mechanism identified on familiar data need not operate under the shift (§6.2).

Generation and selection. Separating generation from selection requires an explicit budget, because accuracy can rise simply through more samples, tokens, verifier calls, or tool interactions. A comparison should either fix this budget or report an accuracy-cost frontier, with all quantities measured under the same shift. Oracle-selection accuracy adds an upper bound on what any verifier could recover from the generated candidates. The gap between the selector actually used and this oracle measures the quality of the selector, and verifier errors on shifted inputs show whether the verifier shares the generator’s shortcuts. Comparing pass@ k with greedy accuracy is also informative for trained models, since post-training may mainly

Primary pillar	≤2023	2024	2025	2026	Total (share)
Training	0	0	13	33	46 (27.2%)
Inference	2	4	5	3	14 (8.3%)
Architecture	1	4	3	13	21 (12.4%)
Analysis	12	16	24	36	88 (52.1%)
Total	15	24	45	85	169 (100%)

Table 1: Core-study counts by primary pillar and first arXiv year. Counts describe the reviewed corpus, not the strength of evidence in each category.

Branch	Core studies
<i>Training (§3) and inference (§4)</i>	
Mid-training	Mehta et al. (2026)
Reinforcement learning	Yuan et al. (2025), Li et al. (2026a), Chaudhry et al. (2026)
Trajectory restructuring	Hou et al. (2026)
<i>Architecture (§5)</i>	
Recurrent depth: weight sharing	Chen (2026), Yu et al. (2026)
Recurrent depth: adaptive computation and stability	Abnar et al. (2023), Kuo et al. (2026), Ibrahimli et al. (2026), Huang et al. (2026a), Labovich (2026), Movahedi et al. (2026)
Information routing: positional encoding and locality	Wu et al. (2026), Shimizu et al. (2026), Madan et al. (2026)
Information routing: modular and neuro-symbolic	Kamali et al. (2025), Shahid and Rothe (2026), Oarga and Du (2025)
Recurrent memory	Xu et al. (2026a), Takashiro et al. (2026)
<i>Analysis (§6)</i>	
Compositional generalization	Saparov and He (2023), Fu and Frank (2024b)
Length, depth, and difficulty	Ding et al. (2024), Stechly et al. (2024), Zheng et al. (2024), Shojaei et al. (2025), Ye et al. (2025a)
Prior exposure	McCoy et al. (2024), Chi et al. (2024), Shelat et al. (2026)
Shared circuits	Zhang et al. (2025c), Yao et al. (2025a)
Learning dynamics	Nanda et al. (2023), Xu (2026), Shi et al. (2026)
Learnability and certification	Guan and Bradic (2026), Zetzsche et al. (2026)

Table 2: Core studies that the main text does not discuss, grouped by the branch of Figure 1 to which we assign them.

raise the probability of an existing solution instead of introducing a new one (Yue et al., 2025). Appendix C.2 defines these metrics.

Generalization matrix. A generalization matrix replaces a single ID/OOD pair with a grid of training and test conditions. Let M_{ij} record performance and inference cost for training condition i and test condition j . Each row contains a matched ID cell, which we treat as its diagonal entry, and each column keeps the task and metric fixed whenever possible (Table 6). Reading along a row shows which shifts an intervention handles, and reading down a column compares interventions on the same test condition. A high diagonal entry shows that an intervention works on the distribution it was designed for, whereas consistent gains off the diagonal show that it transfers beyond the distribution used to build its data, reward, prompt, or controller.

If the loss or metric differs across columns, normalized changes or per-task results are more meaningful than raw gaps.

Asymmetry and interaction. The matrix also exposes two effects that a single OOD score hides. Transfer can be asymmetric: training on a structurally diverse source may help a narrow target even when the reverse direction fails, so both directions should be reported before a favorable pair is taken as a property of the method. Shifts can also interact. Adding single-factor columns and their intersections separates independent gains from interactions, which is what the study of compound shifts requires.

Uncertainty and exposure. Each cell should report uncertainty over test items and, where feasible, over training seeds, prompts, and checkpoints. A gain that appears for one prompt wording or one

Training branch	Transfer mechanism	Strongest current evidence	Main risk
Pre-training	Establish reusable capabilities before adaptation	Controlled training-stage and large-pass@ k comparisons	Opaque exposure makes the boundary hard to identify
Mid-training and curricula	Expose positions, procedures, or domains absent from base training	Long-context and verified-synthesis transfer	More data may repeat the same shortcut
SFT and distillation	Teach explicit operations and reachable traces	Checkpoint sweeps, cross-model transfer, logically isomorphic OOD tests	Teacher forcing, late-stage overfitting, suppressed uncertainty
RL and RLVR	Select and sometimes compose procedures with outcome or process feedback	Held-out domains, capability-controlled pass@ k , composed environments	Reward hacking, entropy collapse, domain interference, dependence on base competence
Hybrid SFT/RL	Combine demonstrations with on-policy exploration	Shared ID/OOD protocol on OpenR1-Math-46k (Table 10)	Gains shrink against correctly implemented SFT→RL; single-run reproductions
Self-improvement and controller-aware training	Learn reusable skills or co-operation with inference controllers	Unseen skill combinations and held-out controller compositions	Specialization to the combinations seen in training

Table 3: Training branches: how each is expected to support transfer, the strongest current evidence, and the main risk.

Inference branch	Helps when	Fails when	Required control
Trajectory restructuring	The task representation is misaligned or intermediate structure is missing	The decomposition is wrong or prompt-sensitive	Matched prompts and paraphrase tests
Compute scaling	Correct paths exist with low but nonzero probability	No useful candidate is generated	pass@ k and an oracle-selection upper bound
State assessment	Errors are easier to detect than to avoid, or difficulty varies across examples	The verifier shares the generator’s shortcut, or the monitor is specific to the source distribution	Independent checks and source-to-target accuracy-cost frontiers
Externalization	Exact operations or reusable skills can be moved outside the model	Tool routing or retrieval fails	Tool-free and retrieval-distance controls

Table 4: Inference branches (§4): when each helps, when it fails, and the control needed to claim transfer.

checkpoint is weaker evidence than a gain that persists across these choices. Exposure audits should be reported separately, since a narrow confidence interval does not rule out overlap with pre-training data.

C Datasets and Evaluation Metrics

This appendix describes the resources behind the evidence in the main text: the supervision a training dataset provides, the shift an evaluation set implements, and the quantity a reported score measures. Tables 8 and 9 list the datasets, and Appendix C.3 documents the comparison of hybrid methods in §3.3.3.

C.1 Datasets

C.1.1 Datasets for Training

Supervision. Training datasets differ in the supervision they provide. Prompts with verifiable

answers support outcome-reward training without demonstrated solutions: DAPO-Math rewrites problems to obtain integer answers, and DeepScaleR, ORZ-Math, and Big-Math enlarge the pool of verifiable prompts (Yu et al., 2025; Luo et al., 2025; Hu et al., 2025a; Albalak et al., 2025). GSM8K and MATH include human-written solutions and can support either supervised or reward-based training (Cobbe et al., 2021; Hendrycks et al., 2021). Demonstration corpora supply reasoning traces to imitate. NuminaMath-CoT contains reformatted solutions, OpenR1 pairs NuminaMath 1.5 problems with several DeepSeek-R1 traces (Li et al., 2024a; Hugging Face, 2025), and LIMO and s1K select small sets of 817 and 1,000 demonstrations, respectively (Ye et al., 2025b; Muennighoff et al., 2025). These categories describe the supervision a dataset offers rather than its only possible use: a trace corpus can supply prompts for RL, and

Field	Minimum report	Misreading prevented
Reasoning task	Generator or dataset, target output, metric, and relevant constraints	Treating a broad benchmark score as evidence about one reasoning process
Reference exposure	Pre-training knowledge if available; all continued-training, SFT, distillation, RL, and adaptation data	Calling an item unseen only because it was absent from the last training stage
Evaluated components	Whether weights, prompts, exemplars, controller, verifier, memory, retriever, and tools remain fixed	Attributing gains from inference methods or external resources to changes in learned model capabilities
Test shift	Named axis, construction method, distance or level, and matched ID control	Combining instance, surface, composition, length, and domain shifts into one OOD label
Outcome and evidence	Performance with uncertainty, generalization gap when comparable, mechanism test if claimed	Treating a probe, score change, or finite test range as stronger evidence than it provides
Inference cost	Samples, generated tokens, tool calls, wall time or FLOPs, and stopping rule	Reporting an accuracy gain that follows only from unbounded extra computation

Table 5: Information required to interpret a reasoning-generalization claim, and the misreading each field prevents.

Training exposure	Matched ID	Surface shift	New composition	Longer/deeper
Base model	score / cost	score / cost	score / cost	score / cost
SFT or distillation	score / cost	score / cost	score / cost	score / cost
RL or hybrid training	score / cost	score / cost	score / cost	score / cost
Fixed-model inference method	score / cost	score / cost	score / cost	score / cost

Table 6: Template for a generalization matrix. Each score should include uncertainty, and each cost should use the same budget definition across columns. The rows can be replaced with the training conditions of a particular study.

a generator can produce supervised traces.

Controllable distributions. Knights and Knaves, Reasoning Gym, SynLogic, and Enigmata provide generators and verifiers with adjustable task parameters (Xie et al., 2025; Stojanovski et al., 2025; Liu et al., 2025a; Chen et al., 2025a), so training and test sets can be separated deliberately by length, structure, or difficulty. A held-out random seed alone yields new instances, not a structural shift. Decontamination serves a different purpose: removing benchmark overlap, as in DeepMath, reduces leakage but does not define an OOD split (He et al., 2025b). Multi-domain resources such as GURU and WebInstruct-verified support source-to-target studies, and DataFlex-RL fixes a mixture of math, logic, and science data to compare data policies (Cheng et al., 2025b; Ma et al., 2025b; Liang et al., 2026). For these resources, domain coverage and verifier reliability should be reported separately.

Counting units. Table 8 counts problems unless stated otherwise, because problems, traces, and task types are not comparable units. OpenR1-Math-220k, for example, contains 225,129 problems in its full release (93,733 in the default configuration

and 131,396 in the extended one), each with two to four traces (Hugging Face, 2025). The hybrid comparisons in §3.3.3 use OpenR1-Math-46k-8192, a subset of the default configuration with correct, short traces (Yan et al., 2025; Limozin et al., 2026).

C.1.2 Datasets for Evaluation

Held-out benchmarks. Mathematical post-training is commonly evaluated on AIME, AMC, MATH-500, Minerva/OCW, and OlympiadBench, with ARC-Challenge, GPQA-Diamond, and MMLU-Pro as checks outside mathematics (Yan et al., 2025; Limozin et al., 2026). We refer to these as the mathematics and out-of-domain suites, although even the mathematical tests may differ from the training data in difficulty, subject, or source. SuperGPQA, HLE, ZebraLogic, BBH, HumanEval, and LiveCodeBench broaden the range of content (Du et al., 2025; Phan et al., 2025; Lin et al., 2025; Suzgun et al., 2023; Chen et al., 2021; Jain et al., 2024). Whether a benchmark is ID or OOD depends on the model’s pre-training, its adaptation data, and the task definition, so the label cannot be assigned to the benchmark alone.

Controlled shifts. Controlled benchmarks aim to change one property while holding the others

Shift axis	Train → test	What it probes
Instance	new samples, same generator	statistical generalization
Surface / format	paraphrase, renamed symbols, new notation	invariance to irrelevant form
Composition	new combinations of seen primitives	rule reuse and recombination
Length / depth	longer inputs, more hops, deeper nesting	extrapolation of iteration
Difficulty	easier or harder than training range	transfer across capability regimes
Task / domain	new task family or subject	portability of procedures
Language modality	/ other language, format, image, audio	reasoning vs. interface
Environment tool	/ new APIs, dynamics, rewards	agentic and procedural transfer
Knowledge	new facts, familiar rules	rule use on new content

Table 7: Axes along which reasoning evaluations shift between training and test. When several axes change together, the source of an improvement or failure becomes harder to identify.

fixed. GSM-IC adds irrelevant context, GSM-Plus applies perturbations that may also change difficulty, and GSM-Symbolic samples variants from mathematical templates (Shi et al., 2023; Li et al., 2024b; Mirzadeh et al., 2025). SCAN, COGS, and CFQ hold out lengths or combinations of familiar primitives (Lake and Baroni, 2018; Kim and Linzen, 2020; Keyzers et al., 2020). MathGAP varies proof depth and width, Compositional-ARC holds out geometric compositions, and OMEGA separates greater complexity from new combinations and new strategies (Opedal et al., 2025; Mondorf et al., 2025; Sun et al., 2025). CLRS-30 was developed to probe neural algorithmic executors and is not a standard LLM leaderboard (Veličković et al., 2022). EsoLang-Bench changes the programming language while keeping the algorithmic problems; the scarcity of public data in these languages is a proxy for low exposure, not proof that pre-training never included them (Sharma and Chopra, 2026). General365 derives 1,095 variants from 365 seed problems, so statistical analyses should account for the shared seeds (Liu et al., 2026a).

Recency and contamination. Examination items and rolling benchmarks support temporal checks only when the items postdate the evaluated model’s relevant training cutoff (Jain et al., 2024; White et al., 2024); AIME 2025, for instance, is not automatically unseen for models trained after its release. GSM1K was withheld at its initial publication, which supported its original comparison with GSM8K, but evaluations after its public release require a new exposure audit (Zhang et al., 2024b). Neither a publication date nor exact-match decontamination excludes exposure to equivalent problems. Table 9 therefore reports

the evaluation axis and dataset version of each benchmark instead of a fixed OOD label.

C.2 Evaluation Metrics for Reasoning Generalization

Reliability under a fixed decoding protocol.

For N problems and a task-specific correctness checker V_q , single-sample accuracy is

$$\text{Acc} = \frac{1}{N} \sum_{q=1}^N V_q(\hat{y}_q).$$

The checker may normalize mathematical expressions, execute code, or compare labels. With k independently sampled responses per problem,

$$\text{avg}@k = \frac{1}{N} \sum_q \frac{1}{k} \sum_{s=1}^k V_q(\hat{y}_{q,s}).$$

This quantity estimates the expected single-sample accuracy under the chosen decoding settings; it does not measure success within k attempts. Maj@ k instead grades the most frequent normalized answer under a stated tie-breaking rule (Wang et al., 2023), so it evaluates an aggregation procedure that spends k samples at deployment.

Coverage at a stated budget. Pass@ k measures whether at least one of k samples succeeds. Given $n \geq k$ independent samples at fixed decoding settings, of which c_q are correct, the estimator is (Chen et al., 2021)

$$\text{pass}@k = \frac{1}{N} \sum_q \left[1 - \frac{\binom{n-c_q}{k}}{\binom{n}{k}} \right],$$

with $\binom{n-c_q}{k} = 0$ when $n - c_q < k$. Because pass@ k relies on an oracle success criterion, it does

not measure the accuracy of a deployable selector unless a suitable verifier is available. A higher pass@1 with unchanged or lower large- k coverage is consistent with post-training that reweights existing solutions (Yue et al., 2025), although finite sampling cannot establish with certainty whether a solution is present or absent. We therefore recommend reporting coverage curves together with sample and token budgets.

Robustness to a matched transformation. For comparable reference and shifted sets, both scores should be reported together with the absolute and relative drops,

$$\begin{aligned}\Delta &= \text{Acc}_{\text{ref}} - \text{Acc}_{\text{shift}}, \\ \Delta_{\text{rel}} &= \Delta / \text{Acc}_{\text{ref}},\end{aligned}$$

where the relative drop requires positive reference accuracy. Under the same 0–1 loss and evaluation procedure, Δ equals the transfer gap defined in §2. A smaller gap is not necessarily better, since it may result from lower reference accuracy. When variants share an original problem or template, results are best aggregated at that level. GSM-Symbolic reports variation across generated test sets (Mirzadeh et al., 2025), and GSM-IC also reports the fraction of base problems solved under *all* distractor variants (Shi et al., 2023). Conditional variant accuracy, $P(\text{variant correct} \mid \text{original correct})$, measures how often correctness is retained on the originally solved subset, but it is not a complete measure of consistency.

Length and composition. We recommend reporting length results as curves that extend beyond the training frontier and that distinguish input length, required depth, and the budget of generated tokens. As a summary, we define the largest tested length up to which accuracy stays at or above a threshold τ :

$$L_{\tau}^* = \max \left\{ L \in \mathcal{L} : \min_{\ell \in \mathcal{L}, \ell \leq L} \text{Acc}(\ell) \geq \tau \right\}.$$

Here $\mathcal{L}$ is the declared grid of test lengths, and an empty set means that no tested length reaches the threshold. The summary says nothing about untested lengths. The ratio $L_{\tau}^* / L_{\text{train}}$ describes extrapolation relative to the final training frontier, similar to the length ratios reported in integer-addition studies (Zhou et al., 2024d). For composition, held-out structures should be compared at matched depth or length where possible (Park et al., 2025).

Transfer and retention. Transfer is best reported as a source-to-target matrix with a base-model reference for every target (Appendix B.2). For method m trained on source s , we define

$$T_{s \rightarrow t}^{(m)} = \text{Acc}_t(\theta_s^{(m)}) - \text{Acc}_t(\theta_0).$$

A positive value on a target excluded from adaptation supports transfer, and a negative value records degradation. Sequential retention is a different quantity: it compares an earlier and a later checkpoint on domains that were already learned. Raw accuracy differences across domains also reflect task difficulty and should not be read as a normalized measure of transfer.

Aggregation, uncertainty, and cost. A domain-balanced summary averages within domains before averaging across them:

$$S_{\text{DB}} = \frac{1}{K} \sum_{d=1}^K \frac{1}{|\mathcal{B}_d|} \sum_{b \in \mathcal{B}_d} \text{Acc}_b.$$

This average assumes comparable units, so rubric credit, F1, and accuracy should not be averaged without a justified normalization. Benchmark coverage alone can reverse method rankings (Liang et al., 2026), which is why per-task scores should accompany any aggregate. Paired method differences need uncertainty estimates over matched training seeds; item-level bootstrap tests, such as those reported for LUFFY, do not capture training-seed variance (Yan et al., 2025). Bootstrap resampling of generated variants should operate on their shared problem or template. Run-level standard deviations and confidence intervals answer different questions and need to be labeled accordingly, and inference methods are compared at a fixed budget or through accuracy-cost curves (Appendix B.2; Snell et al., 2024).

C.3 Protocols and Published Results

Common reproduction study. Table 10 is taken from Limozin et al. (2026), in which the corrected Qwen2.5-Math-7B baselines and the LUFFY and ReLIFT reproductions all use OpenR1-Math-46k-8192. AIME24, AIME25, and AMC are scored with avg@32 and the remaining tests with pass@1, using a temperature of 0.6, a maximum response length of 8,192, and shuffled options for multiple-choice questions. The corrected baselines come from three independent end-to-end runs, whereas

each reproduction is a single run. Sequential training consists of SFT followed by 500 RL steps, and the 50-step variant also raises the RL learning rate from 10^{-6} to 5×10^{-6} . The comparison therefore shares an evaluation protocol but does not match compute.

Published results. Table 11 keeps the originally reported hybrid results separate from the reproduction, and Table 12 summarizes how each method combines SFT and RL. LUFFY and Prefix-RFT report the full nine-benchmark suite (Yan et al., 2025; Huang et al., 2025d). ReLIFT uses avg@8 on MATH-500 and OlympiadBench and does not report Minerva (Ma et al., 2025a), SRFT omits AIME25 (Fu et al., 2025), and HPT reports only two OOD tests, omitting MMLU-Pro, with top- p sampling (Lv et al., 2025). We leave these cells empty rather than importing values from later compiled tables. The original comparisons also differ in SFT implementation and hyperparameters, as audited by Limozin et al. (2026), so the gap between published and reproduced scores mixes implementation differences with seed variance.

Table 8: Summary of datasets used to train reasoning models. † Counts refer to problems, not traces; a problem may have several traces. ‡ Procedural generator; task counts are not instance counts. § Correct, short-trace subset of the OpenR1 default configuration. “~” marks rounded counts.

Dataset	Year	Domain	Problems / Tasks	Supervision
<i>Math: Verifiable Prompts for RL</i>				
GSM8K (train) (Cobbe et al., 2021)	2021	Grade-school math	7,473	Answer + rationale
MATH (train) (Hendrycks et al., 2021)	2021	Competition math	7,500	Answer + solution
SimpleRL-Zoo (Zeng et al., 2025b)	2025	Math	~8K per split	Answer
DAPO-Math-17k (Yu et al., 2025)	2025	Competition math	~17K	Integer answer
DeepScaleR (Luo et al., 2025)	2025	Competition math	~40K	Answer
DeepMath-103K (He et al., 2025b)	2025	Math	103K	Answer + traces
Skywork-OR1 (He et al., 2025a)	2025	Math / code	~105K / 13.7K	Answer / tests
ORZ-Math (Hu et al., 2025a)	2025	Competition math	~129K	Answer
Big-Math (Albalak et al., 2025)	2025	Math	>250K	Answer
<i>Math: Demonstrations for SFT, Distillation, and Hybrid Training</i>				
NuminaMath-CoT (Li et al., 2024a)	2024	Math	~860K	CoT solution
OpenR1-Math-220k (all)† (Hugging Face, 2025)	2025	Math	~225K	2–4 R1 traces per problem
OpenR1-Math-46k-8192†§ (Yan et al., 2025)	2025	Math	~46K	Long CoT (R1) + answer
LIMO (released set)† (Ye et al., 2025b; GAIR, 2025)	2025	Math	817	Long CoT
s1K† (Muennighoff et al., 2025)	2025	Math, science	1,000	Long CoT
OpenThoughts releases† (Guha et al., 2025)	2025	Math, code, science	114K / 1M / 1.2M	Long CoT
<i>Logic, Puzzles, and Procedural Environments</i>				
Knights & Knaves† (Xie et al., 2025)	2025	Logic puzzles	Procedural	Assignment checker
Reasoning Gym† (Stojanovski et al., 2025)	2025	Algorithmic, logic	100+ generators	Task verifier
SynLogic† (Liu et al., 2025a)	2025	Logic	35 task types	Task verifier
Enigmata† (Chen et al., 2025a)	2025	Puzzles	36 task types	Task verifier
<i>Multi-domain</i>				
SciQ (train) (Welbl et al., 2017)	2017	Science MCQ	11,679	Correct option
GURU (Cheng et al., 2025b)	2025	Six domains	92K	Domain verifiers
WebInstruct-verified (Ma et al., 2025b)	2025	Multi-domain	~230K	Model-based verifier
DataFlex-RL mixture (Liang et al., 2026)	2026	Math, logic, science	15K	Rule-based checks

Table 9: Summary of datasets used to evaluate reasoning generalization. The first three groups are general benchmarks ordered by content, and the subsequent groups are built to control a shift or test specific inference/agent environments. Year identifies the referenced release or benchmark version; Compositional-ARC counts meta-learning episodes. “—” denotes not applicable or not reported. * Temporal check: valid only when the model predates the selected items. † Subset used by common harnesses: AMC has 40 or 83 problems, and OlympiadBench commonly uses 675 English text-only mathematics problems from its 8,476-problem release; SRFT lists 674. ‡ Templates $\times$ instances per variant. ¶ Initially withheld; later public evaluation requires an exposure audit. § CLRS counts are per algorithm for node/edge outputs; graph-level outputs use $64\times$ more trajectories.

Dataset	Year	Evaluation Axis	Training	Testing	Eval. Metric
<i>Mathematics</i>					
GSM8K (Cobbe et al., 2021)	2021	Instance	7,473	1,319	Accuracy
Minerva / OCW (Lewkowycz et al., 2022)	2022	Task / domain	—	272	Accuracy
AMC	2022–23	Instance	—	40 / 83 [†]	avg@ k
MATH-500 (Lightman et al., 2024)	2023	Instance	—	500	Accuracy
AIME 2024	2024	Instance	—	30	avg@ k
OlympiadBench (He et al., 2024a)	2024	Difficulty	—	675 [†]	Accuracy
AIME 2025*	2025	Instance	—	30	avg@ k
<i>Science and Knowledge</i>					
ARC-Challenge (Clark et al., 2018)	2018	Task / domain	1,119	1,172	Accuracy
GPQA-Diamond (Rein et al., 2024)	2023	Task / domain	—	198	Accuracy
MMLU-Pro (Wang et al., 2024b)	2024	Task / domain	—	12,032	Accuracy
SuperGPQA (Du et al., 2025)	2025	Task / domain	—	26,529	Accuracy
Humanity’s Last Exam (Phan et al., 2025)	2025	Difficulty	—	2,500	Accuracy
ScienceArena (Zhao et al., 2026a)	2026	Task / domain	—	13 competitions	Rubric credit
<i>Logic, Puzzles, and Code</i>					
ARC-AGI (Chollet, 2019)	2019	Task / domain	400	400 (public)	Accuracy
HumanEval (Chen et al., 2021)	2021	Task / domain	—	164	pass@ k
BBH (Suzgun et al., 2023)	2022	Task / domain	—	23 tasks	Accuracy
LiveCodeBench* (Jain et al., 2024)	2024	Task / domain	—	Rolling	pass@1
Knights & Knaves (Xie et al., 2025)	2025	Difficulty	Procedural	Procedural	Accuracy
ZebraLogic (Lin et al., 2025)	2025	Difficulty	—	1,000	Accuracy
<i>Surface Perturbation and Distractors</i>					
GSM-IC (Shi et al., 2023)	2023	Distractors	—	58,052	Micro / macro acc.
GSM-Plus (Li et al., 2024b)	2024	Perturbation / difficulty	—	10,552	Accuracy
GSM-Symbolic (Mirzadeh et al., 2025)	2024	Surface / format	—	100 $\times$ 50 [‡]	Mean $\pm$ std
<i>Composition and Length</i>					
SCAN (length split) (Lake and Baroni, 2018)	2017	Length / depth	16,990	3,920	Exact match
CFQ (Keysers et al., 2020)	2019	Composition	239,357 in total	—	Exact match
COGS (Kim and Linzen, 2020)	2020	Composition	24,155	21,000	Exact match
Last-letter concatenation (Zhou et al., 2023)	2022	Length / depth	—	500 per length	Accuracy
CLRS-30 [§] (Velićković et al., 2022)	2022	Size / length	1,000	32	Micro-F1
MathGAP (Opedal et al., 2025)	2024	Length / depth	—	Generated	Accuracy
Multi-digit arithmetic (Zhou et al., 2024d)	2024	Length / depth	≤ 10 digits	1,000 digits	Accuracy
Compositional-ARC (Mondorf et al., 2025)	2025	Composition	82,908	8,546	Accuracy
OMEGA (Sun et al., 2025)	2025	Difficulty; composition	Generated	Generated	Accuracy
XDomainBench (Zhiren et al., 2026)	2026	Composition	—	8,598 sessions	Recall / F1
<i>Planning, Agents, and Algorithmic/Modality Shifts</i>					
A-OKVQA (Schwenk et al., 2022)	2022	Modality (Vision)	17,056	1,100	Exact match / Acc
FACTOR (wiki/news) (Muhlgay et al., 2024)	2024	Knowledge shift	—	Generated	Accuracy
WebArena (Zhou et al., 2024c)	2024	Environment / Tool	—	812	Success Rate
Mind2Web (Deng et al., 2023)	2023	Domain / Environment	—	2,000+	Success Rate
SWE-Bench-Verified (Jimenez et al., 2024)	2024	Task / Length	—	500	Code-fixing pass rate
Reversal Curse (Berglund et al., 2024)	2024	Algorithmic direction	Generated	Generated	Exact string-match
<i>Recent and Contamination-limited Benchmarks</i>					
GSM1K* (Zhang et al., 2024b)	2024	Instance	—	1,205 [¶]	Accuracy
LiveBench* (White et al., 2024)	2024	Instance	—	Monthly	Accuracy
General365 (Liu et al., 2026a)	2026	Task; variants	—	365 + 1,095	Accuracy
EsoLang-Bench (Sharma and Chopra, 2026)	2026	Programming language	—	80 per language	Accuracy

Table 10: Accuracy (%) from Limozin et al. (2026); the originally published results are given in Table 11. All rows use Qwen2.5-Math-7B and OpenR1-Math-46k-8192. Corrected baselines report mean $\pm$ standard deviation over three independent runs; mixed-policy reproductions use one run. Math avg. covers six benchmarks and OOD avg. covers the three displayed benchmarks. Training compute is not matched across all rows.

Method	Math avg.	ARC-C	GPQA-D	MMLU-Pro	OOD avg.
Corrected SFT	52.2 $\pm$ 0.2	82.5 $\pm$ 0.6	24.6 $\pm$ 2.9	50.2 $\pm$ 0.3	52.4 $\pm$ 1.1
SFT $\rightarrow$ RL (50 steps)	55.6 $\pm$ 0.5	84.0 $\pm$ 0.5	39.6 $\pm$ 1.0	53.9 $\pm$ 0.4	59.2 $\pm$ 0.2
SFT $\rightarrow$ RL (500 steps)	57.0 $\pm$ 0.3	84.0 $\pm$ 0.7	40.6 $\pm$ 1.8	55.1 $\pm$ 0.6	59.9 $\pm$ 0.9
LUFFY	46.3	81.4	41.8	50.9	58.0
ReLIFT	48.8	80.7	36.7	51.3	56.2

Table 11: Originally reported hybrid scores (%). A shared model and nominal dataset do not imply identical evaluation or compute. ReLIFT uses avg@8 on two mathematical tests; SRFT and HPT omit different tests. Dashes denote absent original results. Means cover the complete six-math or three-OOD suite; ReLIFT’s three-test OOD mean is calculated here from its Tables 1 and 5.

Method	AIME24	AIME25	AMC	MATH-500	Minerva	Olympiad	Math avg.	ARC-C	GPQA-D	MMLU-Pro	OOD avg.
LUFFY (Yan et al., 2025)	29.4	23.1	65.6	87.6	37.5	57.2	50.1	80.5	39.9	53.0	57.8
ReLIFT (Ma et al., 2025a)	28.3	22.9	65.1	87.9	—	57.3	—	81.6	43.1	53.9	59.5
SRFT (Fu et al., 2025)	35.3	—	74.3	89.8	39.7	58.3	—	85.3	46.4	55.9	62.5
Prefix-RFT (Huang et al., 2025d)	31.8	26.4	68.2	88.4	40.3	55.7	51.8	84.0	39.1	52.1	58.4
HPT (Lv et al., 2025)	33.0	21.9	69.4	89.2	46.0	56.9	52.7	81.6	42.9	—	—

Table 12: Settings and limits of the hybrid comparison. All listed methods use Qwen2.5-Math-7B and the filtered OpenR1 data, but training allocations and evaluation details vary.

Method	How SFT and RL are combined	Evaluation or comparison limit
Corrected SFT	Demonstration likelihood; corrected implementation	Three independent runs
SFT $\rightarrow$ RL	Separate SFT, then 500 GRPO steps	Three runs; extra SFT compute
SFT $\rightarrow$ RL (50 steps)	Separate SFT, then 50 GRPO steps	Higher RL learning rate; three runs
LUFFY	Demonstrations in rollout groups	Published or one-run reproduced results, kept separate
ReLIFT	RL interleaved with SFT on difficult questions	Original avg@8 on two tests; no Minerva
SRFT	Entropy-weighted supervised and RL losses	Original table omits AIME25
Prefix-RFT	On-policy continuation of demonstration prefixes	Original reported results
HPT	SFT/RL routing by rollout success	Two OOD tests; top- p sampling

Table 13: Benchmark inventory specifically for Inference-Time Interventions. ID/OOD status is assigned to an evaluation protocol, not to a benchmark name in isolation.

Benchmark / evaluation family	Papers using it	Role and shift construction in the inspected experiments	Metrics / caveats
Last-letter / list concatenation; SCAN	L2M (Zhou et al., 2023); SKiC (Chen et al., 2024)	Synthetic list-length extrapolation. L2M compares prompted decomposition over longer lists. SKiC evaluates exemplar-sized ID tasks versus harder OOD variants.	Accuracy by length.
GSM8K by solution-step count	L2M (Zhou et al., 2023); SKiC (Chen et al., 2024); standard math evaluations elsewhere	Groups questions by solution step count. L2M uses two-step questions as the reference and evaluates gains on questions with ≥ 3 steps.	Accuracy by step-count bucket.
HaluEval-Summary	DaC (Zhang et al., 2024c)	Hallucination-detection and summary task.	F1, accuracy, precision, recall.
SciFact constructed misinformation variant	DaC (Zhang et al., 2024c)	Author-constructed paragraph-level misinformation variant.	F1, G-Mean, precision, recall.
Tower of Hanoi	MAKER (Meyerson et al., 2025); interactive memory-tool study (Malach et al., 2026)	MAKER tests a 20-disk run given a supplied strategy. Malach evaluates length extrapolation by training at 8 disks and testing at 12.	Exact completion/accuracy and cost estimates.
Addition and multiplication length extrapolation	SKiC (Chen et al., 2024); interactive memory-tool study (Malach et al., 2026)	SKiC compares shorter digit lengths (reference) to longer arithmetic instances. Malach trains up to 5/10 digits and tests up to 1,000 digits.	Accuracy/exact recovery by length.
Dynamic programming: maximum non-adjacent sum	SKiC (Chen et al., 2024)	Demonstrations use shorter input lengths; evaluation extends to longer lengths.	Exact match/accuracy by length.
CommaQA-E compositional split	SKiC (Chen et al., 2024)	Evaluates compositional transfer on unseen compositional questions.	Exact Match.
BBH, T4D, MATH (Self-Discover settings)	Self-Discover (Zhou et al., 2024b)	Evaluates task-level performance and model-family transfer.	Task accuracy.
Competition mathematics: AIME24/25/26, CMIMC25, BRUMO25, AIMO/AMC	CFR (Ma et al., 2026); some math papers also use AIME/AMC subsets	Evaluates general benchmark performance and format robustness on competition math.	Pass@1 averaged over four runs (avg@4), accuracy, token overhead.
MATH / MATH-500	ReST-MCTS* (Zhang et al., 2024a); Snell (Snell et al., 2024); TTEL (Chitale et al., 2026); UPAIR (Yan et al., 2026); ORCA (Zhou et al., 2026); VDS-TTT (Moradi et al., 2025); RL ^V (Sareen et al., 2025); RetrievalPRM (Zhu et al., 2025a); MILES (Tong and Gong, 2026)	Utilized in various evaluation protocols including standard test accuracy, difficulty-stratified scaling, verifier assessment, online calibration, and candidate selection.	Accuracy/Exact Match, pass@k, F1/ArithACC, calibration metrics, and token budgets.
GPQA-Diamond and CEval-Hard	ReST-MCTS* (Zhang et al., 2024a); UPAIR/ORCA (Yan et al., 2026; Zhou et al., 2026); TTEL (Chitale et al., 2026); MILES (Tong and Gong, 2026)	ReST-MCTS* uses them as non-source benchmarks compared to MATH-sourced settings. Other papers use them as target evaluation sets.	Accuracy and pass@k/cost depending on paper.
Science/math transfer: SciGLM, SciBench, SciEval	ReST-MCTS* (Zhang et al., 2024a)	Evaluated as non-source benchmark tests.	Average task score/accuracy.
HumanEval(+), MBPP(+)	Boosting PRM/policy models with MCTS (Chan et al., 2025); TTEL (Chitale et al., 2026)	Code-generation evaluation benchmarks.	Pass@k or benchmark score.
ARC-C, ARC-E, OBQA, CSQA, SIQA, FACTOR-Wiki	Residual-stream trajectory detector (Damirchi et al., 2026)	Evaluates assessment transfer by using ARC-C as a reference detection setup and the other datasets as targets.	AUROC / correct-candidate detection.

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Table 13 (continued)

Benchmark / evaluation family	Papers using it	Role and shift construction in the inspected experiments	Metrics / caveats
s1K-1.1; AIME24; GPQA; MATH-500	Thought Calibration (Wu et al., 2025); related calibration evaluations must be checked per model	s1K-1.1 serves as a reference/train-calibration source, while target benchmarks are evaluated for calibration transfer.	Probe AUROC, risk-controlled halting, task accuracy, step/token savings.
AIME24/25, MATH-500, GPQA-Diamond, LiveCodeBench	UPAIR (Yan et al., 2026); ORCA (Zhou et al., 2026); TTEL (Chitale et al., 2026)	Employed for distinct tasks across papers: selective intervention, online calibration, and targeted regeneration.	Accuracy, tokens, diagnosis/risk statistics, and pass@k.
MATH-500, MATH ² , physics evaluation set	RL ^V (Sareen et al., 2025)	Evaluates target/generalization on harder math and physics sets relative to MATH training.	Candidate-selection and final task accuracy.
GSM8K, MATH, OlympiadBench, Omni-MATH (ProcessBench)	RetrievalPRM (Zhu et al., 2025a)	PRM800K is the training source. Competition-level subsets are evaluated as question-OOD.	ProcessBench F1 and ArithACC for first-error identification.
Math tool-use transfer: GSM8K, SVAMP, MAWPS; HotpotQA/Wikipedia-oriented QA	Chain-of-Abstraction (Gao et al., 2025b)	Evaluates fine-tuned tool-using models on zero-shot/held-out task datasets.	Exact match/accuracy, arithmetic errors, latency.
GSM8K-XL and author-designated OOD math sets (including FuncQA)	TECTON (Alazraki and Rei, 2025)	Evaluates tool selection on math problems featuring magnitude and dataset variation.	Final answer accuracy.
GSM8K, MINTQA, Math/Time, IN3, Intention	SMART / SMARTAgent (Qian et al., 2025)	Evaluates tool-use restraint and transfer across varied task types.	Task accuracy, tool calls, tool overuse, and decision confidence.
Mind2Web, WebArena Multi, SWE-Bench	ReasoningBank (Ouyang et al., 2026)	Memory accumulates from task-stream experience. Evaluates performance across several web and site splits.	Success rate, average interaction steps, token cost, issue resolution.
MATH-500, AIME24/25, GPQA-Diamond, LiveCodeBench (MILES settings)	MILES (Tong and Gong, 2026)	Tests modular sub-instruction selection and measures accuracy-token-cost trade-offs.	Majority-vote accuracy, correct-answer/sub-instruction rates, memory units, accuracy-cost frontier.

Table 14: Metric catalogue for inference-time reasoning evaluation. The same metric name may denote different aggregation protocols across papers; scores should be compared only when model, split, sampling/selection rule, and compute budget are aligned.

Metric	Definition / formula	Construct validity / Interpretation	Use and comparability notes
Accuracy / Exact Match	For N examples, $\text{Acc} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[\hat{y}_i = y_i]$. Exact Match applies the benchmark’s exact answer-matching rule.	Higher is better.	Widely reported. Comparability requires matched parsing, normalization, and identical test subsets.
Exact-recovery accuracy	Fraction of generated trajectories/outputs that exactly recover the target algorithm’s output or trace under the benchmark’s rule.	Higher is better.	Used in length-generalization studies (Malach et al., 2026); distinct from answer-only Exact Match.
pass@ k	For n_i sampled candidates with c_i correct, the estimator is $\widehat{\text{pass@}k} = \frac{1}{N} \sum_{i=1}^N \left(1 - \frac{\binom{n_i - c_i}{k}}{\binom{n_i}{k}} \right)$. Evaluates if at least one generated output is correct.	Higher is better; typically non-decreasing with larger k .	Used in sampling/scaling studies. Meaningful comparison requires matched k , sample counts, selection rules, and token budgets.
avg@ k (conventional)	Mean per-sample correctness: $\frac{1}{N} \sum_{i=1}^N \frac{1}{k} \sum_{j=1}^k \mathbb{1}[\hat{y}_{ij} = y_i]$.	Higher is better.	A common standard definition, though actual usage varies by paper.
Paper-specific avg@4 (CFR)	In CFR, pass@1 is averaged over four independent runs: $\text{avg@4} = \frac{1}{4} \sum_{r=1}^4 \text{pass@1}^{(r)}$.	Higher is better.	Averages across multiple runs rather than multiple candidate answers.
maj@ k / majority-vote accuracy	$\frac{1}{N} \sum_{i=1}^N \mathbb{1}[\text{mode}(\hat{y}_{i1}, \dots, \hat{y}_{ik}) = y_i]$.	Higher is better.	Distinct from pass@ k or Best-of- N . Sensitive to sample count, tie-breaking rules, and response diversity.
Best-of- N score	Generate N candidates, select one with a scorer/verifier, then score the selected answer.	Higher task score is better.	Represents a selection protocol rather than a standalone metric. The selector type (verifier, PRM, reward model) and N must be specified.
F1	$F1 = 2 \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$.	Higher is better.	Used for hallucination/verification. Dependent on positive class definition and averaging method.
ProcessBench F1	The harmonic mean of accuracy on erroneous and correct samples, as defined in RetrievalPRM.	Higher is better for first-error identification.	A domain-specific operational definition distinct from standard classification F1.
ArithACC	A weighted arithmetic mean of accuracy on erroneous and correct samples (defined in RetrievalPRM).	Higher is better.	Relies on the paper’s specific weighting scheme.
AUROC	Area under the ROC curve; the probability that a randomly selected positive instance ranks higher than a negative instance.	Higher is better for discrimination.	Measures ranking/discrimination capacity, not the calibration of predicted probabilities.
Risk / selective stopping statistics	Paper-specific bounds such as stopping risk $R(y_t)$ and finite-sample learn-then-test (LTT) guarantees.	Lower risk is preferable; valid guarantees bound risk at a specified confidence level.	Evaluates halting conditions rather than final-answer accuracy.
Success Rate (SR)	For N tasks, $\text{SR} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[\text{task } i \text{ succeeds}]$.	Higher is better.	Used in web-agent evaluation. Dependent on whether success requires all steps or only final task completion.

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Table 14 (continued)

Metric	Definition / formula	Interpretation	Use and comparability notes
Average Steps / Interaction Cost	$AS = \frac{1}{N} \sum_{i=1}^N \text{Steps}(q_i)$, where one step is a full tool/agent interaction cycle.	Lower is usually more efficient, conditional on maintaining task quality.	Measures agentic efficiency; decreases are meaningful only if accuracy is preserved.
Tool-call count / tool overuse rate	Average tool invocations per query. SMART defines overuse as $O = \alpha\beta$ (fraction answerable without tools $\times$ fraction of those where tools were called).	Lower overuse is better.	Context-dependent; must be evaluated alongside task success rates.
Latency, tokens, memory size, and compute budget	Direct operational cost measurements (e.g., wall-clock time, input/output tokens, model calls).	Lower cost is preferable only at matched task performance.	Best evaluated via accuracy-cost frontiers rather than isolated values.
Absolute generalization gap Δ	$\Delta = Acc_{ID} - Acc_{OOD}$.	Larger positive values indicate a larger ID-to-OOD performance drop.	A synthesis metric for paired, protocol-matched ID/OOD settings.
Relative generalization drop	$\Delta_{rel} = \frac{Acc_{ID} - Acc_{OOD}}{Acc_{ID}}$, when $Acc_{ID} > 0$.	Larger positive values indicate a larger proportional drop.	A synthesis metric evaluated alongside the absolute gap.
Target-setting gain over baseline	$G = \text{Score}_{\text{method}, \text{target}} - \text{Score}_{\text{baseline}, \text{target}}$ under the same target protocol.	Positive values favor the evaluated method on that target.	Represents a method-versus-baseline comparison, distinct from an ID-to-OOD generalization gap.

Table 15: Empirical evidence for 24 inference-time methods. † marks the 13 core studies counted under inference; the 11 unmarked entries are contextual methods excluded from the core counts (Appendix A). OOD labels describe the evidence reported in the inspected sources, not an intrinsic property of a benchmark. **Evidence levels (E):** **E1** (Explicit/Strong OOD evaluation with paired ID baseline), **E2** (Terminology-sensitive shift, e.g., out-of-demonstration rather than out-of-distribution), **E3** (Indirect/weak generalization evidence lacking a controlled ID reference), **E4** (No explicit OOD evaluation setup). **Modifiers:** *qualified* (evidence is strictly bounded to a specific shift axis, e.g., length-only), *provisional* (evaluation suggests transfer but lacks fully controlled isolation).

Family	Method / paper	Evaluation evidence	Reported measurements	Generalization evidence and caveat
Trajectory restructuring	Least-to-Most prompting (L2M) (Zhou et al., 2023) [†]	Last-letter/list concatenation, SCAN length splits, and GSM8K grouped by solution-step count.	Accuracy by list length or step count.	E1, qualified. Evaluates structural extrapolation relative to shorter in-context exemplars. Extrapolation across semantic domains is not assessed.
Trajectory restructuring	Divide-and-Conquer (DaC) prompting (Zhang et al., 2024c)	HaluEval-Summary, a constructed SciFact misinformation variant, and long-integer arithmetic tasks.	Accuracy, F1, precision, recall.	E4. The inspected experiments evaluate task performance without an explicit ID/OOD comparative framework.
Trajectory restructuring	MAKER (Meyerson et al., 2025)	Tower of Hanoi with a supplied strategy, including a 20-disk zero-error system run.	Zero-error completion; projected per-model error and cost estimates.	E4, provisional. Performance is evaluated on scaled tasks; a formalized ID/OOD split is not established.
Trajectory restructuring	SKiC (Chen et al., 2024) [†]	Synthetic structural tasks (last-letter concatenation, arithmetic digit-length extrapolation, dynamic programming) and standard reasoning evaluations (CommaQA-E, MATH).	Accuracy/Exact Match by task size and prompt condition; internal-skill activation rates.	E1, qualified. Demonstrates structural length/depth extrapolation on synthetic settings. Reasoning benchmark evaluations lack a paired ID/OOD split.
Trajectory restructuring	Constraint-First Reasoning (CFR) (Ma et al., 2026)	AIME24/25/26, CMIMC25, BRUMO25, and OlympiadBench.	Pass@1 averaged over four runs (avg@4); accuracy and token overhead.	E3. Demonstrates quantitative gains and format robustness on competition math tasks without a predefined ID/OOD baseline.

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Table 15 (continued)

Family	Method / paper	Evaluation evidence	Reported measurements	Generalization evidence and caveat
Trajectory restructuring	Self-Discover (Zhou et al., 2024b)	BBH, T4D, and MATH, including transfer across model families.	Task-level accuracy gains.	E4. Evaluates task performance and cross-model transfer; a data-distribution OOD test is not established.
Trajectory restructuring	Self-Demos (He et al., 2024b) [†]	Query-aware demonstrations and the OOD-Tool set.	Task accuracy compared to few-shot baselines.	E2/E3, terminology-sensitive. Evaluates out-of-demonstration capabilities; does not explicitly test shifts from the model’s pre-training distribution.
Compute scaling	ReST-MCTS* (Zhang et al., 2024a) [†]	MATH-sourced training and evaluation settings compared to non-source benchmarks (GPQA-Diamond, CEval-Hard).	Accuracy and average scores for PRM-guided MCTS and self-training.	E1* (qualified). Reports comparative gains on non-source benchmarks. Base-model pre-training exposure is unquantified.
Compute scaling	Boosting PRMs with MCTS (Chan et al., 2025)	Math, code, and knowledge benchmarks evaluating MCTS-generated training data and inference scaling (Best-of-N, beam search).	Benchmark scores and scaling trends under sample/search budgets.	E3. Evaluates scaling behavior and generated training data improvements without a controlled ID/OOD test.
Compute scaling	Compute-optimal test-time scaling (Snell et al., 2024)	MATH test set stratified into difficulty bins to evaluate search and sequential revision strategies.	MATH accuracy versus generation/compute budget.	E4. Stratifies evaluation by task difficulty; a formal ID/OOD distribution split is not constructed.
Compute scaling	Test-Time Scaling via Error Localization (TTEL) (Chitale et al., 2026)	MATH-500, AIME24/25, GPQA-Diamond, and LiveCodeBench evaluating localized regeneration.	Pass@k and token budgets under rollout/sample constraints.	E4. Evaluates pass@k improvements under localized revision; explicit train/test distribution shifts are not assessed.
State assessment	Residual-stream trajectory detector (Damirchi et al., 2026) [†]	ARC-C (reference) and other reasoning/commonsense datasets (targets) used for cross-benchmark candidate detection.	AUROC and correct-candidate detection rates.	E1, assessment transfer. Evaluates the transferability of the assessment/detection signal to held-out benchmarks.
State assessment	UPAIR (Yan et al., 2026)	Diagnostic experiments and selective interventions evaluated on MATH-500, AIME24/25, GPQA-Diamond, LiveCodeBench.	Final accuracy/token cost; partial diagnosis and intervention metrics.	E4, provisional. Evaluates diagnostic signals and intervention benefits without a controlled OOD comparison.
State assessment	Thought Calibration (Wu et al., 2025) [†]	Hidden-state probes and risk-controlled halting evaluated on target sets (AIME24, GPQA, MATH-500) using s1K-1.1 as reference data.	Probe AUROC, stopping/risk statistics, task accuracy, step/token savings.	E1, qualified. Assesses calibration and task gains on target sets; calibration stability under distribution shift is measured separately.
State assessment	Online Reasoning Calibration (ORCA) (Zhou et al., 2026) [†]	Test-time online calibration evaluated on MATH-500, AIME24/25, GPQA-Diamond, LiveCodeBench.	Accuracy, generated tokens, and risk/calibration statistics.	E1, qualified. Explicitly evaluates calibration under target settings.
State assessment	Generative verifier / RL ^V (Sareen et al., 2025) [†]	Candidate selection evaluated on MATH500, harder MATH ² , and a physics set following MATH training.	Candidate-selection accuracy under parallel test-time scaling.	E1, task generalization only. Demonstrates downstream selection capabilities on math-adjacent and physics tasks.
State assessment	Verifier-Driven Sample Selection TTT (VDS-TTT) (Moradi et al., 2025)	High-confidence outputs used as pseudo-labels for generator adaptation on GSM8K, MATH-500, and AIME.	Exact Match/accuracy against a frozen base and verifier-only baselines.	E3. Evaluates target task gains from generator adaptation; a paired reference distribution is not defined.

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Table 15 (continued)

Family	Method / paper	Evaluation evidence	Reported measurements	Generalization evidence and caveat
State assessment	Retrieval-Augmented PRM (Retrieval-PRM) (Zhu et al., 2025a) [†]	ProcessBench (GSM8K, MATH, OlympiadBench, Omni-MATH) evaluated using a PRM800K and Math-Shepherd retrieval pool.	ProcessBench F1 and ArithACC for first-error detection.	E2, qualified. Competition subsets are evaluated as question-OOD relative to PRM training.
Inference-time externalization	Interactive tool-use memory for bounded-state models (Malach et al., 2026) [†]	Evaluates interactive memory tools on synthetic addition/multiplication (up to 1,000 digits), logical graphs, and Tower of Hanoi.	Exact-recovery and task accuracy by sequence size.	E1, structural length generalization only. Evaluates structural length generalization for specific architectures; semantic or algorithmic OOD is not tested.
Inference-time externalization	Chain-of-Abstraction (CoA) (Gao et al., 2025b) [†]	Zero-shot evaluations on math datasets (GSM8K, SVAMP, MAWPS) and knowledge-intensive QA.	Exact match/accuracy, arithmetic error rate, latency.	E1, provisional cross-task transfer. Evaluates cross-task transfer on designated target datasets. Dataset overlap with pre-training is not assessed.
Inference-time externalization	TECTON (Alazraki and Rei, 2025) [†]	Tool selection evaluated on GSM8K-XL and designated math sets with magnitude/dataset variation.	Final task accuracy.	E1, provisional. Demonstrates tool selection transfer on math word problems featuring magnitude shifts.
Inference-time externalization	SMART / SMARTA-agent (Qian et al., 2025) [†]	Tool-use decisions evaluated across task types including GSM8K, MINTQA, and Math/Time-style tasks.	Task accuracy, tool calls, tool overuse rate, decision confidence.	E1, provisional. Evaluates tool-use restraint and cross-task transfer.
Inference-time externalization	ReasoningBank (Ouyang et al., 2026)	Task-stream memory accumulation evaluated on Mind2Web, WebArena Multi, and SWE-Bench-style tasks.	Success rate, average interaction steps, token cost.	E3. Demonstrates strategy-level memory and within-stream improvements; an isolated held-out target is not established.
Inference-time externalization	MILES (Tong and Gong, 2026)	Modular sub-instruction selection evaluated on MATH-500, AIME24/25, GPQA-Diamond, LiveCodeBench.	Majority-vote accuracy, sub-instruction rates, memory size, accuracy–cost frontier.	E3 for auxiliary transfer checks; E4 for main results. Evaluates accuracy and cost trade-offs; cross-domain memory transfer is not directly assessed.

D Further Analytical Evidence

This appendix expands each paragraph of §6 under the same heading and adds the expressivity results on which the theoretical paragraphs build. Table 16 summarizes the lines of analysis, and Table 17 lists representative behavioral diagnostics with the shift each one targets.

D.1 Behavioral Evidence

Compositional generalization. The decline reported by Dziri et al. (2023) was measured on multiplication, logic-grid puzzles, and dynamic programming. On formal languages, performance depends on whether a prompt stays within the rules learned in training (Mészáros et al., 2024). Accuracy on MathGAP falls as proof trees become deeper, wider, and nonlinear (Opedal et al., 2025), and XDomainBench reports a collapse as more scientific domains are combined, partly because each added domain also makes the problem harder (Zhiren et al., 2026). Since both benchmarks enlarge the combination together with its depth or difficulty, neither isolates composition; two-hop questions come closer to doing so because the number of steps is fixed. Scale alone does not provide systematicity. On Compositional-ARC, a 5.7M-parameter model meta-trained for compositionality generalizes to unseen combinations of geometric transformations where frontier LLMs do not (Mondorf et al., 2025).

Length, depth, and difficulty. These axes can diverge within a single environment. In a controlled shortest-path setting, models transfer to unseen maps but fail as the planning horizon grows, which the authors attribute to recursive instability (Tong et al., 2026). Length transfer is usually partial. Scratchpads improve extrapolation on algorithmic tasks without removing the underlying failure (Anil et al., 2022), and the seed dependence reported by Zhou et al. (2024d) is consistent with underspecification: the same data admit both an extrapolating solution and a shortcut, and each training run settles on one of them (§6.3). Length generalization can also transfer between tasks, as training on a longer, related auxiliary task helps a target task extrapolate beyond its own training lengths (Cai et al., 2025). For difficulty, ordering deductive-reasoning data from easy to hard gives no robust advantage over random order under either SFT or RL (Mordig et al., 2026).

Prior exposure. In chess, both performance and the benefit of each additional reasoning token decline as relevant priors become sparse (Pleiss et al., 2026). When a hard perturbation invalidates the original solution, models keep applying the memorized technique, and they do so more often when the original problem appears in context (Huang et al., 2025a). Tasks designed to fall outside pre-training, such as programming in esoteric languages, reveal failures that familiar benchmarks hide (Sharma and Chopra, 2026). Knowledge that is present is not always used. Models that cannot reverse a fine-tuned fact succeed when the same fact is given in context (Berglund et al., 2024), and models can acquire abstract structure yet make only limited use of it for compositional generation at test time (Chen et al., 2026b).

Key takeaways

- **Transfer is axis-specific.** Success on one named shift does not predict success on another.
- **Knowing is not composing.** Stored components can fail to combine, even at scale.
- **“Unseen” is stage-relative.** Apparent generality depends on the reference-exposure stage.

D.2 Mechanistic Evidence

Shared circuits. de Varda et al. (2026) localize their circuits with attribution patching. In small transformers, causal interventions can recover compact RASP subprograms that suffice to explain length-generalizing models (Huang et al., 2026b), so reusable computation can be located directly instead of being inferred from scores. Reuse has identifiable limits. Whether pattern matching succeeds on an instance is predicted by how many training contexts witness the relevant functional equivalence, and when a variable reaches the output through several paths, models fail to form a single intermediate representation (Chang et al., 2025). Together with the failure on newly integrated knowledge noted in §6.2 (He et al., 2026), these limits mean that overlap measured on familiar data supports a transfer claim only if it also holds on the shifted split.

Interfaces between steps. In transformers trained from scratch on two-hop queries, lower layers build a representation of the bridge entity, but when the second hop departs from its training

distribution, upper layers map that representation to outputs instead of reasoning over it (Zhang et al., 2026b). For facts injected by fine-tuning, relocating their internal representations by self-patching substantially improves failed cases, which is consistent with the facts being stored but not routed to the layers where later computation uses them (Dai et al., 2026). In small transformers trained from scratch on SCAN, a homomorphism error measured on hidden states predicts compositional transfer under noise injection, and regularizing this error during training improves the transfer (An and Du, 2026). Because the regularizer is a training intervention, this result goes beyond correlation, although it may change more than the measured property. Read alongside the two-hop results in §6.2, these findings locate the bottleneck at the hand-off between operations rather than in the presence of the facts or subskills.

Learning dynamics. With extended training, vanilla transformers come to generalize hierarchically on structurally novel inputs long after in-domain accuracy has saturated (Murty et al., 2023). A grokked rule can nevertheless remain bound to the facts seen in training (Wang et al., 2024a), which mirrors the limit of circuit reuse on new knowledge. Regularization can act within a critical period: on a controlled compositional task, weight decay applied during a single window of training matches the out-of-distribution accuracy of weight decay applied throughout training (Ali, 2026). In LLM post-training, excessive SFT can erode the plasticity that later RL needs (§3.3.3; Liu et al., 2026c). Each of these findings is tied to particular models, seeds, and finite test ranges, and theory is needed to separate such regularities from guarantees.

Key takeaways

- **Overlap predicts transfer.** Shared circuits predict transfer across variants, yet can fail on new knowledge.
- **Failures sit at interfaces.** Computed intermediate results may not reach the next step.
- **Optimization selects the solution.** Training length and the evaluated checkpoint shape what transfers.

D.3 Theoretical Evidence

Expressivity. Learnability results presuppose that the model class can represent the target procedure, and what it can represent depends on the com-

putation budget and the state encoding. The number of intermediate steps is itself a resource. Under assumptions on layer normalization, transformer decoders with linearly many decoding steps recognize all regular languages, and with polynomially many steps they recognize exactly the problems solvable in polynomial time (Merrill and Sabharwal, 2024). Without intermediate steps, constant-depth transformers with constant-bit precision are limited to AC^0 ; with as many chain-of-thought steps as the size of a Boolean circuit, they can evaluate that circuit (Li et al., 2024c). Extrapolating constructions further depend on how state is encoded. A universal-transformer construction provably evaluates Boolean circuits of any length and depth, but it assumes fully parenthesized inputs, a positional encoding that tracks gate depth, and a number of iterations that grows with depth (Ito et al., 2026). Such constructions do not carry over directly to finite-precision LLMs, and representability alone does not show that training will find them.

Learnability. Scratchpads whose content is unsupervised do not break the globality barrier of Abbe et al. (2024), and the vocabulary of the trace matters as well. Under standard positional encodings and a finite alphabet, chain-of-thought learning that generalizes in length cannot go beyond TC^0 , whereas signpost tokens, whose vocabulary grows with problem size, allow a length-generalizable simulation of Turing machines (Kraus et al., 2026). Decomposition can reduce sample complexity. When the learner receives feedback on intermediate states, a recursive curriculum for simulating semiautomata over T steps reduces the required supervision from $\Omega(T)$ to $2^{O(\sqrt{\log T})}$ tokens (Rajaraman et al., 2026). Empirically, ablations in tiny transformers trained on Dyck languages attribute the gain from intermediate traces to their content rather than to the extra tokens (Valle and Reid, 2026). These results suggest why scratchpads give only partial gains under length shift: a trace helps when it turns a global dependency into local, reusable updates, and otherwise it adds tokens without changing what must be learned.

Other results concern inductive bias, which determines which of several fitting solutions training reaches. Beyond the logarithmic-depth construction in §6.3, Liu et al. (2023) show that solvable semiautomata can be simulated with depth independent of T . Zhou et al. (2024a) conjecture,

with experimental support, that transformers length-generalize on tasks that have a short program in RASP-L, a restricted language of transformer-compatible operations. On synthetic state-tracking tasks, [Huang et al. \(2025c\)](#) prove that gradient descent learns chains of thought whose length extrapolation is governed by the algebraic structure of the task, through attention concentration. [Kazemnejad et al. \(2023\)](#) prove that a decoder without positional encoding can represent both absolute and relative positions, and they observe that after training its attention mostly resembles a relative scheme and extrapolates better than explicit encodings. In the retrieval task of [Singh et al. \(2026\)](#), rotary encodings make attention logits depend only on the relative offset, whereas a head with learned absolute encodings stays fixed to a position inside the training range. Data matter as well, but they are not sufficient on their own: the guarantees of [Ahuja and Mansouri \(2024\)](#) for length and compositional transfer require both diverse training support and constraints that rule out direct memorization.

Certification. The negative result in §6.3 concerns C-RASP, a formalism linked to length generalization, so its reach depends on what C-RASP can express; for regular languages, this is determined by their algebraic decomposition ([Yang et al., 2026b](#)). For the positive fragment of C-RASP, which is equivalent to fixed-precision transformers, a computable length generalization bound exists but is necessarily exponential ([Yang et al., 2026a](#)). For causal transformers with learnable absolute positional encodings, [Huang et al. \(2025b\)](#) characterize the functions that an idealized inference scheme with a norm-based regularizer identifies in the limit. This characterization proves that length generalization is possible for a rich family of problems, and in experiments it predicts success and failure across algorithmic and formal-language tasks. The positive result of [Izzo et al. \(2026\)](#) in §6.3 also bounds the training length it requires. Plan verification provides a positive case for checking rather than generating: transformers provably learn to verify long plans in a large class of classical planning domains, even as the vocabulary grows with the number of objects ([Sarraf et al., 2026](#)).

Key takeaways

- **Representable is not learnable.** Expressible procedures need not be learned.
- **Locality and bias govern selection.** Global targets resist learning without local steps, and standard training can prefer brittle shortcuts.
- **Finite tests do not certify.** Accuracy on a finite range is evidence about that range alone.

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Line of analysis	Question answered	Main finding	Interpretation limit
<i>Behavioral evidence (§6.1)</i>			
Compositional generalization	Do familiar primitives transfer to new combinations?	Loss grows with the depth, width, and order of the combination	Template overlap can make recombination look like composition
Length, depth, and difficulty	Does the procedure continue beyond the trained range?	Transfer is partial, axis-specific, and sensitive to the training seed	A finite extrapolation range is not unbounded extrapolation
Prior exposure	How far does the test lie from earlier training exposure?	Benefits shrink as priors become sparse, conventions change, or facts are queried in the unseen direction	Pre-training exposure is inferred, not observed
<i>Mechanistic evidence (§6.2)</i>			
Shared circuits	Do successes on both sides of a shift share a computation?	Circuit overlap predicts format transfer and item-level correctness	Overlap on familiar data need not hold on the shifted split
Interfaces between steps	Can each step read the result of the previous one?	Bridge entities are bypassed, resolved too late, or stored out of reach; homomorphism error predicts transfer	Repairing patches are selected per case; training interventions may change more than the measured property
Learning dynamics	Which of several fitting solutions does training select?	Rules emerge late, can stay bound to seen facts, and depend on regularization timing	Tied to specific models, seeds, and checkpoints
<i>Theoretical evidence (§6.3)</i>			
Learnability	Will training on finite data select a procedure the model class can represent?	Training finds brittle shortcuts unless locality, diverse support, or symmetry favor extrapolating solutions	Representability does not imply learnability; assumptions rarely match LLM training
Certification	Can a finite test establish extrapolation?	Finite training lengths suffice for restricted classes; no computable test length covers all two-layer C-RASP programs	Positive results assume idealized training or cover verification only

Table 16: Lines of analysis: the question each answers, its main finding, and the limit on interpreting it. Rows follow the paragraph headings of §6. Behavioral rows rest on performance under a named shift, mechanistic rows on representational or interventional evidence, and theoretical rows on results under explicit assumptions.

Diagnostic	Primary shift	Main observation	Interpretation limit
GSM-Plus (Li et al., 2024b)	Statements and answer targets	Correctness is brittle even on variants of solved problems	Perturbations remain within grade-school arithmetic
GSM-Symbolic (Mirzadeh et al., 2025)	Numeric instantiation and distractor clauses	Equivalent templates yield unstable accuracy	Template generation cannot test every skill
MathGAP (Opedal et al., 2025)	Proof-tree depth, width, and shape	Accuracy declines with structural complexity	Generated word problems simplify language variation
Compositional-ARC (Mondorf et al., 2025)	Unseen transformation combinations	Meta-learned small models can exceed frontier LLM systematicity	Abstract spatial tasks are controlled but narrow
General365 (Liu et al., 2026a)	Domain and variant	Broad reasoning remains below specialized math performance	Variant construction may not remove familiarity
XDomainBench (Zhiren et al., 2026)	Cross-domain composition	Failure grows with composition order and interaction	Knowledge and reasoning are partly entangled
Numeric remapping (Barker et al., 2026)	Surface-preserving numeric change	Robustness differs sharply by dataset structure	Arithmetic word problems only
Elenchos (Steiglechner et al., 2026)	Abductive rule mutation	Detection is stronger than causal attribution	Formal systems are controlled but narrow
OOD visual planning (Neuhaus et al., 2026)	Map size and representation	CoT helps ID more than most OOD settings	Simple navigation does not span open-world planning
Chess generalization (Pleiss et al., 2026)	Density of useful priors	Performance falls as relevant priors become sparse	Training exposure is inferred, not observed
OMEGA (Sun et al., 2025)	Mathematical novelty type	Separates exploratory, compositional, and transformative transfer	Novelty taxonomy depends on task construction
CoT as a mirage (Zhao et al., 2025)	Data distribution	Apparent reasoning depends on distribution design	Behavioral evidence does not identify a mechanism

Table 17: Representative behavioral diagnostics: the shift each targets, its main observation, and the limit on interpreting it.